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RESEARCH PAPERS

December-2025

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TITLE:

Comprehensive Review on Rehabilitation Strategies for Indian Bridge Infrastructure

AUTHOR :

D.; Shelake, A.G.; Minde, P.R.

JOURNAL NAME:

Journal of Structural Design and Construction Practice (Vol.-31, Issue-2)

DETAILS:

Published on 23 December 2025

ABSTRACT:

This paper underscores the critical role of bridges in India's transportation network and highlights the pressing need for effective rehabilitation strategies to address their aging infrastructure. With over 200 bridges exceeding 75 years of age, the transportation network urgently requires proactive rehabilitation efforts. While government initiatives like Setu Bharatam aim to tackle this challenge, the current approaches remain inadequate and largely reactive. This review delves into various rehabilitation strategies, including repair, replacement, retrofitting, and reconstruction, examining their materials, methodologies, and specific applications. It emphasizes that the selection of an appropriate strategy should be tailored to the unique defects and conditions of each bridge. By categorizing common defects in aging bridges and aligning them with suitable rehabilitation methods, the study offers a systematic framework to enhance the safety, durability, and longevity of bridge infrastructure. Furthermore, a comparative analysis of materials and techniques highlights the strengths and limitations of each approach, advocating for a structured and proactive strategy that prioritizes cost-effectiveness and safety. The paper concludes by providing recommendations for the optimal application of rehabilitation methods, addressing the specific challenges faced in the Indian context.

Link: https://doi.org/10.1061/JSDCCC.SCENG-1947

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Influence of Aggregate Gradation and Mineral Admixtures on the Properties of Permeable Concrete: An Experimental Study

AUTHOR :

Sathe, S.; Ansari, U.; Razvi, S.W.N.; Kangda, M.Z.

JOURNAL NAME:

Journal of Structural Design and Construction Practice (Vol.-31, Issue-2)

DETAILS:

Published on 23 December 2025

ABSTRACT:

Strength and permeability are crucial design factors for permeable concrete (PC), but limited research has examined the influence of aggregate size and mineral admixtures on PC properties. In the studies, the constant head permeability method was used to find the coefficient of permeability of the PC, as there is no standard method for measuring it correctly and precisely in laboratory conditions. X-ray fluorescence spectroscopy, scanning electron microscopy, and Fourier-transform infrared spectroscopy analyses were performed on cement, fly ash (FA), and microsilica (MS) to characterize their morphology, chemical properties, quality, and reactivity. Also, the experiment aims to investigate the influence of various aggregate grading (single-, two-, and three-particle size gradations), MS and FA, and water-to-binder ratio (0.20, 0.25, and 0.30) with a constant C/A ratio of 0.25 on the compressive strength (CS), split tensile strength (STS), flexural strength (FS), permeability, and porosity of PC. The findings demonstrate that three-particle size gradation mixes give higher CS, STS, and FS than single- and two-particle size gradations. For single-sized aggregates, permeability decreases with increasing particle size. A combination of three-particle size gradations results in the lowest permeability. The maximum CS and maximum permeability for the combined use of MS and FA are found at the replacement levels of 5% and 10%, respectively. The investigation also established the correlation between porosity and permeability.

Link: https://doi.org/10.1061/JSDCCC.SCENG-2016

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TITLE:

Design, synthesis, antifungal activity, and molecular docking studies of benzothiazole, S-benzyl-2,4-isodithiobiuret, and thiourea derivatives of 1-hepta-O-benzoyl-β-D-maltose nanoparticles

AUTHOR :

Zyate, S.C.; Awajare, N.V.; Gaikwad, S.S.; Raut, K.; Jha, V.; Jadhao, A.R.; Agrawal, P.T.

JOURNAL NAME:

Bioorganic and Medicinal Chemistry Letters
(Vil,-132)

DETAILS:

Published on December 2025

ABSTRACT:

The surge in fungal infection-related mortality worldwide is being caused by drug resistance to well-established antifungals such azole derivatives (bifonazole, fluconazole, miconazole, and clotrimazole). Finding novel molecules that are structurally different to these could be a useful tactic to get over medication resistance that is currently on the market. In an effort to develop highly potent non-resistance antifungal agents, we reported a series of compounds with benzothiazole, S-benzyl-2,4-isodithiobiuret and thiourea derivatives of 1-hepta-O-benzoyl-β-d-maltose NPs and their antifungal activity against the most infectious fungal strain Candida albicans. Numerous analogues among the synthesized compounds have shown potent antifungal activity. All the synthesized compounds were tested in vitro for determining their anticandidal activity. Almost all the compounds were found to be highly potent than established antifungal drugs (MIC ¼ 0.25–0.125 mg mL−1) against Candida albicans strain. An in silico molecular docking study was also performed to comprehend the mode of action of the active compounds towards prospective target 1EA1 binding protein.

Link: https://doi.org/10.1016/j.bmcl.2025.130499

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TITLE:

High-performance Na3V2(PO4)3-based asymmetric sodium-ion full cells: Progress, strategies, and future outlook

AUTHOR :

Kate, R.; Chothe, U.; Deokate, R.; Kalubarme, R.; Chavali, M.; Kale, B.

JOURNAL NAME:

Journal of Power Sources (Vol.-665)

DETAILS:

Published on December 2025

ABSTRACT:

This review article highlights the growing importance of sodium-ion batteries (SIBs) as a promising alternative to lithium-ion batteries (LIBs) due to the abundance and cost-effectiveness of sodium. However, to meet the demands of commercial applications, it is essential to study sodium-ion full cells (SIFCs). Among the various cathode materials, Na3V2(PO4)3 (NVP) stands out for its stable sodium superionic conductor (NASICON) structure, high ionic conductivity, and excellent electrochemical properties. To advance the development of NVP-based SIFCs, this review provides a comprehensive analysis of the current status of NVP SIFCs based on different types of anodes, focusing on their electrochemical characteristics in full-cell configurations. The review also summarizes various modification strategies to improve electrochemical performance, including presodiation, electrolyte component refinement, and optimizing the capacity ratio. These strategies can enhance the overall performance of NVP-based SIFCs and bring them closer to commercial viability. Finally, the review provides valuable insights and directions for future research to improve NVP-based SIFCs. By addressing the main challenges and limitations associated with these batteries, researchers can pave the way for more efficient and cost-effective energy storage solutions that meet the growing demands of the energy sector.

Link: https://doi.org/10.1016/j.jpowsour.2025.239010

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Injectable thermosensitive hydrogel with green synthesized metallic nanoparticles for sustained Withaferin-A delivery in breast cancer

AUTHOR :

Srinivasan, M.; Bhavsar, D.; Ahiwale, R.; Chandane, A.Y.; Pawar, A.; Tagalpallewar, A.; Baheti, A.M.

JOURNAL NAME:

Journal of Drug Delivery Science and Technology (Vol.-116)

DETAILS:

Published on December 2025

ABSTRACT:

Thermosensitive copolymers exhibit controlled drug release through temperature-induced sol-to-gel transitions and micelle dynamics. This work aims to develop an in situ gelling hybrid hydrogel system (HHS) incorporating PLGA (75:25) and PEG with metallic nanoparticles (silver and zinc oxide) to enhance the therapeutic efficacy of withaferin-A (WA) in combating breast cancer. The hydrogel was characterized utilizing 1H NMR, FT-IR, and rheological behavior. The nanoparticles were developed via sustainable green synthesis using Camelia sinensis extract and characterized using FT-IR, zeta potential, XRD, TEM, and FE-SEM. Silver and zinc oxide nanoparticles exhibited a zeta potential of −17.9 mV and −12.9 mV, and particle sizes of 255 nm and 128.5 nm, respectively. XRD affirmed crystallinity with 2θ values of 39.1° and 38.2° for silver and zinc oxide NPs, respectively. TEM and FE-SEM disclosed spherical morphology for AgNPs and cylindrical morphology for ZnO NPs. In vitro cytotoxicity studies on MCF-7 and MDA-MB-231 cell lines demonstrated that WA-entrapped silver or ZnO NPs in hydrogels reduced cell viability to 38 % and 37 %, respectively. The calculated IC50 values of AgNPs and Zn NPs were 14.87 μg/mL and 18.87–19.49 μg/mL, respectively, indicating enhanced potency. Flow cytometry confirmed a total apoptosis rate of ~87 % for WA-loaded formulations, comparable to paclitaxel, with statistical significance ($p = 0.03$ and $p < 0.01$). Drug release obeyed Peppas kinetics with R2 values up to 0.92, signifying controlled diffusion. The hydrogel exhibited suitable properties and sustained release for localized breast cancer therapy.

Link: https://doi.org/10.1016/j.jddst.2025.107907

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TITLE:

Improved link net-based energy prediction: Clustering-based routing and improved kernel-LMS algorithm for data aggregation in WSN

AUTHOR :

Gawali, V.S.; Pande, M.; Sayyad, M.; Bhadade, R.S.

JOURNAL NAME:

Computers and Electrical Engineering (Vol.-130)

DETAILS:

Published on December 2025

ABSTRACT:

Advancements in Wireless Sensor Networks (WSNs) are critical to enabling smart computing applications. These networks are composed of compact, battery-powered sensor nodes that operate under strict energy and resource constraints. Such limitations often lead to uneven energy consumption, adversely affecting network lifespan. Although clustering techniques help reduce energy usage, conventional methods frequently fail to ensure balanced energy distribution among nodes. Hence, this research presents the Secretary Bird Merged Walrus Optimization (SBMWO) method for effective cluster head (CH) selection, taking into account a number of constraints such link lifetime, risk, delay, energy prediction, and distance in order to handle these issues. An Improved LinkNet-based Transmission Model (ILNTM) is employed to predict the residual energy of nodes, aiding in optimal CH selection and enhancing network longevity. Following CH selection, an Improved Kernel-based Least Mean Square Algorithm (IK-LMSA) is used for data aggregation, effectively eliminating redundant data and improving both energy efficiency and network lifetime. Experimental results show that the proposed SBMWO method achieves a competitive mean performance score of 0.962, outperforming several existing methods. These outcomes confirm the effectiveness of the proposed framework in enhancing energy prediction accuracy, data aggregation efficiency, and network stability in WSNs.

Link: https://doi.org/10.1016/j.compeleceng.2025.110855

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TITLE:

Enhancing corrosion resistance: a comprehensive analysis of surface treatments for steel in washdown environments

AUTHOR :

Patil, S.; S.T.; Jawade, S.

JOURNAL NAME:

International Journal of Materials Engineering Innovation (Vol.-17, Issue-1)

DETAILS:

Published on 22 December 2025

ABSTRACT:

Steel is a preferred material for mechanical components due to its optimal strength, ductility, and wear resistance, making it vital for bearings, couplings, and gears. Stainless steel, with its high chromium content, offers excellent corrosion resistance but may lack the strength needed for high-stress applications. In the food and beverage industry, frequent use of chloride and sulphide cleaners, coupled with high operating stresses, accelerates corrosion in components like bearings and gears. While stainless steel resists corrosion, it is susceptible under these conditions, necessitating materials that combine high strength with superior corrosion resistance. This paper examines the surface corrosion of stainless steel and proposes surface treatments to enhance its strength and corrosion resistance. It explores the role of alloying elements in improving corrosion resistance and evaluates surface treatment methods to incorporate these elements, addressing the industry's demand for robust, corrosion-resistant materials.

Link: https://doi.org/10.1504/IJMATEI.2026.150696

TITLE:

Influence of CFRP strip width and fibre orientation on the shear and flexural performance of RC beams

AUTHOR :

Gaidhankar, D.; Mahajan, H.G.; Sathe, S.; Gunaware, P.D.; Ingle, G.

JOURNAL NAME:

Innovative Infrastructure Solutions (Vol.-11, Issue-1)

DETAILS:

Published on 4 December 2025

ABSTRACT:

Carbon fibre-reinforced polymers (CFRP) are widely used to reinforce concrete due to their superior corrosion resistance, durability, and flexibility compared to conventional steel reinforcement. This study aims to investigate the structural performance of reinforced concrete (RC) beams strengthened with CFRP sheets, focusing on the influence of fibre orientation and width on shear and flexural behaviour. Six RC beams with dimensions of 1500 mm × 200 mm × 150 mm were cast and externally strengthened using CFRP U-jackets at a 45° orientation, identified as the most effective configuration compared to 0° and 90°. The specimens were tested under two-point loading to evaluate load-carrying capacity, deflection behaviour, ductility index, energy absorption and failure modes. Furthermore, finite element analysis (FEA) was performed to numerically model and validate the experimental findings. The experimental and FEA results showed strong agreement, confirming the reliability and accuracy of the numerical model. Beams strengthened with wider CFRP U-jackets exhibited notably higher load-carrying capacity, reduced deflection, and delayed crack propagation. Overall, the study demonstrates that optimising the orientation and width of CFRP sheets can significantly improve the strength and ductility of RC beams, providing a reliable and durable solution for structural rehabilitation and retrofitting.

Link: https://doi.org/10.1007/s41062-025-02405-z

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TITLE:

A Comprehensive Approach to Retrofittable ADAS for Heavy Vehicle

AUTHOR :

Kulkarni, Y.; Chavan, P.; Salke, P.; Kute, R.; Gohokar, V.

JOURNAL NAME:

Lecture Notes in Networks and Systems (1507 LNNS)

DETAILS:

Published on 23 November 2025

ABSTRACT:

This paper presents an innovative retrofittable advanced driver assistance system (ADAS) that is designed specifically to address the issue of heavy-vehicle blind spots in high-volume traffic conditions. The novel approach uses the DATS_2022 dataset containing Indian traffic scenarios and deploys a YOLOv5 small model on a Jetson Nano developer kit optimized to support real-time object detection. The research specifically addresses the challenges posed by right-hand drive configurations in India, with emphasis on the blind spots towards the left and back sides of the vehicle. It employs a dynamic view-switching functionality in the graphical user interface (GUI), which automatically switches views with alerts to enhance visibility of the left and back blind spots, crucial for right-hand drive configuration of heavy vehicles. The design emphasizes high frame-per-second (FPS) performance, data stream with low latency, as well as depth estimation for alert generation. Furthermore, it comes with a bespoke user interface with a wired interface and an Android App for wireless connection. The YOLOV5 model has achieved an accuracy of 80% while running at an average FPS of 12 on a Jetson Nano 4 GB developer kit.

Link: https://doi.org/10.1007/978-981-96-8750-3_25

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TITLE:

Analysis of Explainable AI Techniques in a Computer Vision-Based Maritime Surveillance System

AUTHOR :

Thengade, A.; Kale, P.; Ghorpade-Aher, J.; Gunjal, A.

JOURNAL NAME:

Lecture Notes in Networks and Systems (Vol.- 1540 LNNS)

DETAILS:

Published on 20 November 2025

ABSTRACT:

In recent years, Explainable Artificial Intelligence (xAI) has emerged as a key area of interest in deep learning application, with a surge in inquiries, articles, and conferences. The integration of AI into various domains, particularly defense and surveillance, has been driven by advancements in computational photography and AI methods in Computer Vision. However, the reliance on black-box deep learning models without interpretability has led to challenges in finding explanations for their outputs. Modern surveillance systems using object detection and image classification algorithms often misjudge objects, resulting in increased false positives due to external factors. These black-box models lack justification or interpretation, making them unreliable for autonomous operation. To tackle this challenge, xAI techniques have been used for generating explanations/interpretations as to why a certain model classifies or detects certain objects. In this research endeavor, we introduce the xAI approaches utilized in a critical domain, i.e., Maritime Defense and Surveillance with four pre-trained Computer Vision models. Our key observation is that Local Interpretable Modelagnosic Explanations (LIME) provide good result in pinpointing the features contributing the most in the prediction. We intend to address why a model misclassifies certain images/objects (caused by color gradient differences around the area of interest) to make them more robust using qualitative predictions.

Link: https://doi.org/10.1007/978-981-96-9533-1_20

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Understanding Unsupervised Learning Using Hierarchical Clustering

AUTHOR :

Pabalkar, V.; Bokhare, A.; Lenka, R.M.; Chitranshi, J.

JOURNAL NAME:

Lecture Notes in Networks and Systems (Vol.- 1654 LNNS)

DETAILS:

Published on November 2025

ABSTRACT:

Crime is one of the most worrying and widespread issues of our society. Criminal deterrence is essential for people safety. The overall crime inference when assessed, does help to keep a record of crime and assist in avoiding adversities. The aim of the study is to examine patterns in data acquired over time. Criminal violations offend humanity, and it should be prosecuted as soon as possible. Criminology is the scientific method of understanding crime and the motives behind the act. Criminology is an interdisciplinary area which gathers data and conducts further study into such offenses. While there is such a large amount of data on criminal activities, identifying and preventing crimes is one of the most difficult tasks. It is imperative to develop approaches and procedures for predicting future crimes and taking appropriate preventative steps. Cluster analysis includes breaking down huge data to minute groups with similar or identical characteristics. We can evaluate and assess methods, structures, layouts and interactions that are present in the data using visualization tools, so as to help uncover interesting areas and acceptable parameters for future analysis.

Link: https://doi.org/10.1007/978-3-032-06697-8_10

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TITLE:

Empowering SMEs with Blockchain: Advancing Sustainability in Water Systems

AUTHOR :

Bapat, G.; Pujari, P.; Kumar, A.; Eslamian, S.

JOURNAL NAME:

Blockchain Technology for Water and Environmental Systems (Book Chapter)

DETAILS:

Published on November 2025

ABSTRACT:

As small and medium-sized enterprises (SMEs) grapple with the imperative of sustainability in the context of water and environmental systems, blockchain technology emerges as a transformative ally. This chapter delves into the pivotal role blockchain plays in bolstering SMEs' efforts toward sustainability. It explores the unique challenges SMEs face in adopting eco-friendly practices and outlines the fundamental principles of blockchain technology that make it an ideal solution. With a spotlight on applications, this chapter showcases how blockchain enhances supply chain transparency, aids in water management, tracks carbon footprints, facilitates renewable energy trading, and fosters circular economy initiatives for SMEs. Real-world case studies exemplify successful integrations of blockchain in SMEs, revealing tangible benefits in economic, social, and environmental spheres. While illuminating the promise of blockchain, the chapter also addresses potential challenges and considers future trends. In a world increasingly concerned with sustainability, this exploration elucidates how blockchain can empower SMEs to be drivers of positive change, forging a path toward a more sustainable future for water and environmental systems.

Link: https://doi.org/10.1201/9781003599593-16

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।। विश्वशान्तिर्धुवं ध्रुवा ।।

TITLE:

A layered and highly porous poly-ionic liquid-based membrane as a gel polymer electrolyte for rechargeable lithium-ion batteries

AUTHOR :

Potdar, A.S.; Chothe, U.; Chavali, M.; Kale, B.; Kulkarni, M.V.

JOURNAL NAME:

Journal of Materials Chemistry A (Vol.-13, Isssue-48)

DETAILS:

Published on 13 November 2025

ABSTRACT:

Gel electrolytes have great significance in solid-state batteries; in this regard, lithium-ion conducting gel polymer electrolyte (GPE) membranes were developed using an optimized blend of the polymeric ionic liquid poly(diallyldimethylammonium)bis(trifluoromethanesulfonyl)imide (PDADMATFSI), poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP), and lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) salt. The resulting GPEs were systematically characterized for their morphological and electrochemical properties. The morphological study of the GPE membrane clearly shows a unique layered structure with distinct layers. The optimized membrane exhibited a high ionic conductivity of 1.48×10^{-3} S cm^{-1} at 25 °C, which increased to 3.17×10^{-3} S cm^{-1} at 65 °C, indicating efficient lithium-ion transport. It also demonstrated excellent mechanical flexibility, a broad electrochemical stability window of 4.8 V, and a lithium-ion transference number of 0.75. When employed in a Li|GPIL-3|$LiFePO_4$ half-cell configuration, the membrane enabled a high initial discharge capacity of 127.09 mAh g^{-1} at 0.1C. which increased to 155.75 mAh g^{-1} by the fifth cycle. Moreover, it delivered 151 mAh g^{-1} at 1C with a capacity retention of 72.91% over 600 cycles, highlighting its robust rate capability and cycling stability. Ionic transport in the GPE membrane occurs primarily within the amorphous regions, where mobile Li^+ cations coordinate with electronegative atoms along the polymer backbone, enabling stepwise migration between transient coordination sites and facilitating efficient conduction. Additionally, the membrane's layered structure is responsible for ion transport. Notably, the ionic conductivity achieved at room temperature surpasses previously reported values for PDADMA-based GPE systems. These findings underscore the potential of the optimized GPE composition for high-performance lithium-ion batteries.

Link: https://doi.org/10.1039/D5TA06822C

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TITLE:

Creating GPU Kernels for Expression Trees Generated at Runtime with RandomArt

AUTHOR :

Kulkarni, S.

JOURNAL NAME:

Conference: SA Technical Communications '25: SIGGRAPH Asia 2025 Technical Communications

DETAILS:

Published on 14 December 2025

ABSTRACT:

Procedurally generated art often relies on the evaluation of mathematical expression trees, where the structure of the tree defines a function over pixel coordinates. The system introduced in this paper generates such expression trees at runtime using a Probabilistic Context-Free Grammar (PCFG), seeded for reproducible randomness. These large, heap-allocated trees are computationally expensive to evaluate on CPUs, especially at higher image resolutions. To overcome this, the system compiles runtime-generated trees into GPU kernels using Apple's Metal framework, enabling efficient procedural image generation for deep trees and large images. This work details the system's modular architecture, PCFG-driven tree synthesis, Metal-based code generation for GPU execution, and performance implications. Empirical results show that GPU execution achieves up to 1910 × speedup compared to closure-tree interpretation and 72 × compared to a Cranelift JIT backend, highlighting the substantial advantage of GPU specialization for runtime generative art.

Link: https://doi.org/10.1145/3757376.3771407

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TITLE:

Riverbank Filtration Facilitated the Attenuation of Organic Micropollutants in the Floodplains of the River Yamuna, India

AUTHOR :

Mishra, S.; Kumar, R.; Kumar, P.; Mehrotra, I.; Bonasia, R.; Mora, A.; J.; Kumar, M.
Santosh
.

JOURNAL NAME:

ACS ES and T Water (Vol.-5, Issue-12)

DETAILS:

Published on November 2025

ABSTRACT:

We investigated the occurrence of organic micropollutants (OMPs) in the floodplains of the River Yamuna, Delhi, India, with a primary focus on corroborating the fate and transport of OMPs in highly pollution-stressed floodplains being tapped for potable water supply. In addition, we traced the seasonality of riverbank filtrations (RBFs) related to the aggravation or attenuation of contaminants of emerging concern. Fifty-seven OMPs were identified, quantified, and categorized into pharmaceutically active compounds (PhACs), pesticides, personal care products (PCPs), phthalates, endocrine-disrupting chemicals (EDCs), fatty acids, food additives, hormones, and hospital wastes. Ranney wells (RWs) exhibited the presence of a lower number of OMPs with attenuated concentrations with respect to the surface water. Largely, RBF/sand filtration appears to be an effective, low-cost pretreatment step, naturally offered as the first step in potable water supply to densely populated areas. We believe that the efforts made to tap the RWs for a potable water supply in Delhi, rather than deep groundwater, provide a fair degree of sustainability to the entire process. However, a fully developed and precisely operated wastewater treatment system can be a potential way to keep the RWs safe for a longer period, providing a clean, safe, and potable water supply.

Link: https://doi.org/10.1021/acsestwater.5c00383

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TITLE:

Distributed Iterative Multi-k-mer Metagenome Assembly with Progressive Graph Simplification and GC-Aware Clustering

AUTHOR :

G.D.; W.; Das, A.K.

JOURNAL NAME:

Conference: BCB '25: Proceedings of the 16th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics

DETAILS:

Published on 10 December 2025

ABSTRACT:

This poster presents a prototype of our distributed metagenome assembly pipeline, which combines multi-k-mer progressive assembly, iterative graph simplification, and GC-aware clustering to improve contig quality and scalability. Using Dask for parallelization, the pipeline processes FASTQ reads through paired-end parsing, De Bruijn graph construction, and targeted subgraph refinement. On a 16S rRNA dataset (SRR2628505), it outperformed MEGAHIT with a higher N50 (651 bp vs. 609 bp) and 1.5x faster runtime, showing promise for larger datasets.

Link: https://doi.org/10.1145/3765612.3767768

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TITLE:

Shifting perceptions, building confidence: a social marketing approach to smoking cessation through Decisional Balance and Self-efficacy

AUTHOR :

Teeli, A.A.; Singh, J.; Pal, D.

JOURNAL NAME:

Journal of Social Marketing (Vol.-15, Issue-4)

DETAILS:

Published on 5 August 2025

ABSTRACT:

Purpose This study aims to examine the theoretical workings of the transtheoretical model (TTM) of behavior change and its application in social marketers' smoking cessation interventions. Design/methodology/approach Data analysis was conducted to explore the mediating roles of Decisional Balance and Self-efficacy within the TTM framework, focusing on their influence on different stages of change (Pre-contemplation, Contemplation, Action and Maintenance) in smoking cessation. Established measurement tools, including URICA, the Decisional Balance Scale, Processes of Change Questionnaire and the Smoking Abstinence Self-efficacy Scale, were used to collect data, which was analyzed using Structural Equation Modeling and squared multiple correlations in AMOS 25. Findings The study found that Decisional Balance mediates the influence of Experiential Processes on the Pre-contemplation and Contemplation stages. In addition, Self-efficacy mediates the impact of Behavioral Processes on the Action and Maintenance stages. Practical implications The findings suggest that social marketing interventions can be tailored more effectively based on the smoker's Stage of Change. This targeted approach can enhance the efficacy of interventions designed to prevent smoking and assist smokers in quitting. Originality/value This research provides novel insights into the mediating roles of Decisional Balance and Self-efficacy within the TTM in the context of smoking cessation. It offers valuable guidance for developing more effective social marketing strategies tailored to different stages of smoking cessation, an area not extensively covered in previous studies. The study splits the TTM into two halves, discerning the relevance of each half for pre-cessation and post-cessation stages.

Link: https://doi.org/10.1108/JSOCM-05-2024-0121

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TITLE:

Enhanced Photoelectrochemical Performance of BiVO4 Photoanodes Through Few-Layer MoS2 Composite Formation for Efficient Water Oxidation

AUTHOR :

Patil, D.R.; Patil, S.S.; Mishra, R.K.; Mane, S.M.; Ryu, S.Y.

JOURNAL NAME:

Materials (Vol.-18, Issue-24)

DETAILS:

Published on 15 December 2025

ABSTRACT:

Photoelectrochemical water splitting (PEC-WS) provides a sustainable route to transform solar energy into hydrogen; however, its overall efficiency is constrained by the inherently slow kinetics of the oxygen evolution reaction. Bismuth vanadate (BiVO4) is considered an attractive visible-light-responsive photoanode due to its suitable band gap (~2.4 eV) and chemical stability; however, its efficiency is restricted by limited charge transport and significant charge carrier recombination. To overcome these limitations, BiVO4–MoS2 (BVO–MS) heterostructures were synthesized through a simple in situ hydrothermal approach, ensuring robust interfacial coupling and uniform dispersion of MS nanosheets over BVO dendritic surfaces. This intimate contact promotes rapid charge transfer and improved light-harvesting capability. Structural and spectroscopic analyses confirmed the formation of monoclinic BVO with uniformly integrated amorphous MS. The optimized BVO–MS10 electrode delivered a photocurrent density of 4.72 mA cm^{-2} at 0.6 V vs. SCE, approximately 5.3 times higher than pristine BVO, and achieved an applied bias photon-to-current efficiency of 0.49%. Mott–Schottky analysis revealed a distinct negative shift in the flat-band potential for BVO–MS10, indicative of an upward movement of its conduction band and the establishment of a strong internal electric field that enhances charge separation and interfacial electron transport. These synergistic effects collectively endow the in situ engineered BVO–MS heterostructure with superior PEC water oxidation performance and highlight its promise for efficient solar-driven hydrogen generation.

Link: https://doi.org/10.3390/ma18245639

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Artificial inteligence and datasets for leukemia diagnosis: A scoping review of machine lerning and deep learning approaches

AUTHOR :

A.; Mane, R.

JOURNAL NAME:

MethodsX (Vol.-15)

DETAILS:

Published on December 2025

ABSTRACT:

Leukemia is the cancerous disease of the blood and the bone marrow that causes excessive proliferation of abnormal white blood cells, if detected too late, it can lead to potentially fatal consequences. Peripheral blood smear examinations and bone marrow biopsy are example of conventional diagnostic techniques that are invasive, time-consuming and subject to human variability. Recent advances in artificial intelligence (AI) particularly in the areas of Machine Learning (ML) and Deep Learning (DL)offer encouraging answers by making it possible to detect and classify leukemia using automated, effective and precise techniques.

With an emphasis on image-based techniques based on publicly available datasets such as ALL-IDB, Csingle bondNMC, AML_Cytomorphology_LMU, SN-AM and CPTAC-AML, this paper reviews the most recent research on Machine Learning and Deep Learning approaches includes Convolutional Neural Networks (CNNS), ResNet, DenseNet, MobileNet and ensemble models for leukemia diagnosis. The survey highlights some of the most significant issues, such as dataset imbalance, stain variability, lack of standard annotations and limited clinical validation. The paper also discusses research gap and future initiatives including Explainable AI, lightweight deployment models, clinically reliable diagnostic system and hierarchical classification framework aligned with WHO 2022 classification standards.

Link: https://doi.org/10.1016/j.mex.2025.103722

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Scientific Research Methodology: Principles, Tools, and Techniques

AUTHOR :

S.S.; S.G; A.K.

JOURNAL NAME:

Book: Scientific Research Methodology: Principles, Tools, and Techniques

DETAILS:

Published on December 2025

ABSTRACT:

This book serves as a comprehensive overview of the principles, tools, and techniques of scientific research methodology. It guides readers through various research methodologies and techniques, starting from the basics and culminating in scientific writing and publication. Using simple language, this well-rounded resource empowers readers of all levels – from beginners to seasoned experts – to publish their research outcomes effectively. Some of the key features are as follows:

Offers a thorough exploration of research methodologies, literature survey strategies, and classification of research approaches

Explains fabrication and manufacturing techniques and advanced characterization methods to bolster key concepts and practical skills

Emphasizes data optimization and post-processing techniques and introduces state-of-the-art tools for effective analysis and interpretation of data, including a brief introduction to MATLAB, Python, machine learning, and artificial intelligence

Discusses documentation and scientific writing, offering invaluable advice for creating and publishing impactful scientific manuscripts Covering all essential steps necessary for conducting technical research, this book is particularly tailored to benefit new researchers in science and engineering fields engaged in academic studies and technical writing.

Link: https://doi.org/10.1201/9781003664871

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Numerical analysis of surface settlement with saturated zone during urban tunnelling

AUTHOR :

Ingle, G.; Banne, S.; Gaidhankar, D.; K.; Aswar, D.

JOURNAL NAME:

Research on Engineering Structures and Materials (Vol.-11, Issue-6)

DETAILS:

Published on December 2025

ABSTRACT:

Surface settlement assessment during urban tunnelling, especially in the presence of saturated bodies, is crucial for the safety of nearby structures and utilities. This study uses a finite element-based Rocscience (RS2) tool to obtain the surface settlement trough for the two cases, viz. tunnel hits a water pocket (saturated body) and grouted water pocket. In the case of tunnel hits a water pocket, the maximum surface settlement was found to be 31.6 mm, while for the grouted water pocket, it was 16.4 mm, which is nearly 50% less than the latter case. Furthermore, the support capacity curve findings showed that the support lining is unstable when the tunnel hits a water pocket; however, in the grouted water pocket case, the liner's safety exceeds the design factor of safety. In addition, the combined maximum surface settlement obtained by the numerical method (RS2) is validated with the O'Reilly and New method, based on Peck's classical empirical theory. It is found to be closer to it by 6%, indicating good agreement. The findings are also compared with a few analytical studies, including Limanov's, Sagaseta's and the Gonzales-Sagaseta method. Limanov's and Sagaseta's, methods showed comparable results to the present study; however, the Gonzales-Sagaseta method underestimated the results. Finally, the applicability of the RS2 method is verified by conducting the parametric study on tunnel geometry (depth and diameter), and the results compared to analytical methods. The application of predicted surface settlement guides the tunnel engineer from the future consequences regarding the destructive effects of adjacent structures and facilities, particularly in the presence of water bodies.

Link: http://dx.doi.org/10.17515/resm2025-571st1219%20

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Vorinostat polymeric nanoparticle gel: A promising therapy for management of psoriasis

AUTHOR :

Kaur, T.; Hinge, N.; Pukale, S.; Ansari, M.N.; Upadhyay, J.

JOURNAL NAME:

Journal of Drug Delivery Science and Technology (Vol.-114)

DETAILS:

Published on December 2025

ABSTRACT:

The present study aimed to develop and evaluate a PLGA-based nanoparticles (PNPs) gel of vorinostat to enhance its therapeutic efficacy in the treatment of psoriasis. Vorinostat-loaded PNPs were prepared using the nanoprecipitation method using minimum amount of solvent and surfactant. The nanoparticles were characterized for particle size, polydispersity index (PDI), zeta potential, and entrapment efficiency, while the vorinostat-loaded PNPs gel was evaluated for rheological properties, in vitro drug release, skin permeation, and retention. The optimized PNPs exhibited a mean particle size of 161.70 ± 1.32 nm, PDI of 0.16 ± 0.03, zeta potential of −7.53 ± 0.04 mV, and entrapment efficiency of 89.87 ± 3.28 %. The vorinostat-loaded PNPs gel showed sustained drug release (88.05 ± 0.51 % over 72 h) and a five-fold increase in skin retention (16.82 ± 1.26 μg/cm2) compared to plain vorinostat gel. In vivo evaluation using an imiquimod-induced psoriatic mouse model demonstrated significant improvement in PASI scores, histopathological features, and immunohistochemical markers compared to the marketed formulation and plain vorinostat gel. These findings suggest that vorinostat-loaded PNPs gel offers a promising, scalable, and safer topical strategy for enhanced dermal delivery and effective management of psoriasis.

Link: https://doi.org/10.1016/j.jddst.2025.107492

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Evolution of Drying Process Modeling: From Mathematical Foundations to Machine Learning-Driven Prediction and Control

AUTHOR :

Talele, P.; Chaudhari, V.D.; Talele, H.; D.V.

JOURNAL NAME:

Journal of Food Process Engineering (Vol.-48, Issue-12)

DETAILS:

Published on 19 December 2025

ABSTRACT:

Modeling the drying process is essential for optimizing efficiency, reducing energy consumption, and improving product quality in food and agricultural systems. Over time, modeling approaches have progressed from empirical and semi-empirical correlations to theoretical and diffusion-based models, which offer insight into transport mechanisms but are often limited in adaptability and real-time prediction. Computational and multiphysics frameworks have enhanced accuracy by coupling heat, mass, and momentum transfer phenomena; however, their high computational cost restricts widespread industrial use. The advent of machine learning (ML) has transformed drying analysis by enabling data-driven, adaptive, and real-time modeling. Techniques such as artificial neural networks (ANN), support vector machines (SVM), random forest (RF), adaptive neuro-fuzzy inference systems (ANFIS), and hybrid models provide superior flexibility in handling complex, nonlinear, and multivariate drying data. This review presents a comprehensive overview of the evolution of drying process modeling—spanning empirical, theoretical, computational, and ML-based approaches—across four critical domains: (i) drying kinetics and moisture prediction, (ii) product quality evaluation, (iii) energy and exergy analysis, and (iv) process control and automation. The synthesis highlights the transition toward intelligent, data-driven drying systems with improved prediction accuracy, optimized energy use, and enhanced product quality.

Link: https://doi.org/10.1111/jfpe.70300

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Intelligent multimodal 3D biometric recognition using PointNet + + for robust face–ear authentication

AUTHOR :

Kaur, V.; Bhatt, D.P.; Tharewal, S.; Tiwari, P.K.

JOURNAL NAME:

Scientific Reports (Vol.-15, Issue-1)

DETAILS:

Published on November 2025

ABSTRACT:

In real-world applications, the reliability of biometric recognition systems that are based on 2D modalities is typically reduced due to limitations such as sensitivity to changes in illumination, facial expressions, and occlusion among other things. To overcome these problems, this research offers a multimodal biometric model that incorporates data from 3D face and 3D ear to achieve reliable identity recognition. The 3D biometrics offer more comprehensive structural information than the 2D characteristics, and they are more resistant to the effects of environmental changes. These 3D features are then used to recognize and secure storage of multimodal biometrics. Initially pre-processing steps, including cropping, normalization, hole filling, and spike removal are applied on 3d biometrics. After that, feature extraction is performed using the PointNet++model, which is a network based on Convolutional Neural Networks (CNNs) that processes point clouds directly. We used the Face Recognition Grand Challenge (FRGC) database for 3D images of faces and the University of Notre Dame (UND) Collection G database for 3D images of ears for the tests. Our tests show that the PointNet++model is accurate 99% of the time for 3D face recognition and 98% of the time for 3D ear recognition. With its 3D point cloud optimization and resilient architecture, the PointNet++model achieves high accuracy for 3D face and 3D ear by learning multi-scale features that capture both local and global information.

Link: https://doi.org/10.1038/s41598-025-27757-5

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

First instance of non-sexual cannibalism in Plexippus paykulli (Audouin) (Araneae: Salticidae) in India

AUTHOR :

Patra, A.; Kuni, N.; Koparde, P.

JOURNAL NAME:

Israel Journal of Entomology (Vol.-54)

DETAILS:

Published on December 2025

ABSTRACT:

Plexippus paykulli is a dimorphic salticid. Males possess median white stripes on a black carapace and abdomen, whereas females are brownish, with a white me-dial band on the thoracic region and a white median band accompanied by two subdorsal white spots on the posterior third of the abdomen. Plexippus paykulli is widely pantropical and usually inhabits bushes, shrubs and human settlements. On 25 May 2021, near Bara Solemanpur Village (21.67210°N 87.57483°E) in Purba Medinipur District, West Bengal, India, the authors observed an unusual instance of cannibalism in which a male P. paykulli was seen consuming a conspecific male. The predator male was noticeably larger than the prey male, and the event lasted for about five minutes.

Link: https://doi.org/10.5281/zenodo.17787293

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Enhancing software fault prediction through data balancing techniques and machine learning

AUTHOR :

Raj, A.; Chavan, D.M.; Agarwal, P.; Gigi, J.; Rao, M.; Musale, V.; Chanchlani, A.; Dholkawala, M.S.; Kumar, K.V.

JOURNAL NAME:

IAES International Journal of Artificial Intelligence (Vol.-14, Issue-6)

DETAILS:

Published on December 2025

ABSTRACT:

Software fault prediction is essential for ensuring the reliability and quality of software systems by identifying potential defects early in the development lifecycle. However, the presence of imbalanced datasets poses a significant challenge to the effectiveness of fault prediction models. In this paper, we investigate the impact of different data balancing techniques, including generative adversarial networks (GANs), synthetic minority over-sampling technique (SMOTE), and NearMiss, on machine learning (ML) model performance for software fault prediction. Through a comparative analysis across multiple datasets commonly used in software engineering research, we evaluate the efficacy of these techniques in addressing class imbalance and improving predictive accuracy. Our findings provide insights into the most effective approaches for handling imbalanced data in software fault prediction tasks, thereby advancing the state-of-the-art in software engineering research and practice. An extensive experimentation is performed and analyzed in this study here that includes 8 datasets, 4 data balancing techniques, and 4 ML techniques in order to demonstrate the efficacy of various models in software fault prediction.

Link: http://doi.org/10.11591/ijai.v14.i6.pp4787-4801

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Implementation of Syzygium cumini Garden Waste Biochar as a Sustainable Source for the Synthesis of Paints and Pigments

AUTHOR :

Hanwatkar, G.R.; Kale, G.R.; Joshi, R.R.

JOURNAL NAME:

Nature Environment and Pollution Technology (Vol.-24, Issue-4)

DETAILS:

Published on December 2025

ABSTRACT:

With increasing concerns about environmental degradation and the need for sustainable alternatives, the repurposing of organic waste materials, such as garden waste biochar, is considered a feasible opportunity to mitigate waste accumulation by creating value-added products. The present research study aims to identify the potential and investigate the feasibility of utilizing Syzygium cumini garden waste biochar as a sustainable source for pigment and paint formation. Biochar was produced from garden waste via carbonization, highlighting its chemical composition and physical properties that are conducive to the development of pigments and paints. Furthermore, this study examined various extraction techniques to isolate pigments from biochar by assessing their color properties, stability, and compatibility with paint formulations. These findings suggest that Syzygium cumini garden waste biochar holds promise as a sustainable source for pigment and paint formation, offering environmental benefits and contributing to the circular economy. In the case of feedderived pigments, the Syzygium cumini biochar-based pigment exhibited a comparatively better color intensity. Spectrophotometric analysis indicated high color retention and stability over time, suggesting its viability for various applications in the paint and dye industries. This establishes the novelty of this work, as commercial pigments are oil-based products that generate carbon footprints. This study contributes to the growing body of knowledge on sustainable material utilization and provides insights for the development of eco-friendly paint products in the construction and coatings industries.

Link: https://doi.org/10.46488/NEPT.2025.v24i04.B4307

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Occlusion-aware dual-memory transformer for unsupervised UAV video stabilization with adaptive inference paths

AUTHOR :

Sapkal, R.J.; Warhade, K.K.

JOURNAL NAME:

Signal, Image and Video Processing (Vol.-19, Issue-17)

DETAILS:

Published on December 2025

ABSTRACT:

bilizing unmanned aerial vehicle (UAV) videos remains challenging due to severe jitter, occlusion, and nonlinear scene motion. Classical trajectory-smoothing methods often assume rigid motion and fail under parallax and occlusion, while recent learning-based stabilizers such as DeepStab and DUT apply fixed pipelines that ignore frame-level variations in jitter and visibility. Transformer-based frameworks have improved temporal modeling, yet they still lack explicit occlusion reasoning and adaptive computation, limiting robustness and efficiency in dynamic UAV environments. We propose the Occlusion-Aware Dual-Memory Transformer (OADMT), an unsupervised framework that separates motion and occlusion cues via dual memory branches. An occlusion-gated attention mechanism suppresses unreliable regions, while a multi-resolution decoder and uncertainty-guided inpainting enhance stabilization quality. An adaptive inference path selection module dynamically switches between lightweight and full-capacity branches based on jitter and occlusion scores, balancing accuracy with computational cost. Experiments on UAV123-OS, DeepStab, and synthetic UAV datasets demonstrate that OADMT reduces inter-frame jitter by 17.6%, improves PSNR by 12.3%, and lowers inference time by 21.4% compared with DeepStab and other recent stabilizers, while maintaining SSIM above 93% in cross-domain tests. These results establish OADMT as an efficient and occlusion-robust stabilizer well suited for real-time UAV applications.

Link: https://doi.org/10.1007/s11760-025-04993-w

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Investigation of reduction in refrigerant charge for domestic split air conditioner

AUTHOR :

Phadake, H.C.; Mali, K.V.

JOURNAL NAME:

International Journal of Air-Conditioning and Refrigeration (Vol.-33, Issue-1)

DETAILS:

Published on 27 November 2025

ABSTRACT:

This study analyses the vapor compression cycle for the reduction of refrigerant charge without affecting the cooling capacity and COP. Globally, use of natural refrigerants is increasing. HC-290 is a possible refrigerant. HC-290 is highly flammable and is classified under A-3 safety class. Charge reduction below the safe operating limit is the requirement. The condenser carries majority of the charge in the vapour compression cycle. (VCC) Improvement in condenser internal volume by the change in flow direction by modifying the condenser orientation indicates a potential method for charge reduction. Analysis is carried out for a falling film condenser (FFC) for the use of HC-290 refrigerant in split type air conditioner (STAC) of cooling capacity of 5000 W complying the flammability limit as per international standard. Analytical calculations for the FFC and the CFD simulation for the falling film condensation has verified charge reduction. An experimental set up with same condenser tube was built. Number of condenser tubes in service were different compared to standard STAC condenser. At different operating conditions cooling capacity was monitored. The VCC process parameters were measured and cooling capacity, power consumption and coefficient of performance (COP) were calculated. When the 25% condenser tubes were not in service compared to all tubes in service, cooling capacity reduced by 2% compared to baseline cooling capacity. The energy consumed was marginally reduced so higher COP was observed. The charge reduction of 45 g is observed with near to no compromise on cooling capacity.

Link: https://doi.org/10.1007/s44189-025-00086-y

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Stability, release mechanisms, and applications of double emulsions loaded with Opuntia extracts: a comprehensive review

AUTHOR :

Soni, Y.; Bashir, O.; Morya, S.; Mondal, I.H.; Pawase, P.A.; Mugabi, R.

JOURNAL NAME:

Beni-Suef University Journal of Basic and Applied Sciences (Vol.-14, Issue-1)

DETAILS:

Published on 28 November 2025

ABSTRACT:

Background: Opuntia cactus, or prickly pear, is a good source of bioactive compounds, especially betalains that have high antioxidant activity and play an important role in human health. However, despite their nutritional and functional benefits, these compounds are highly sensitive to environmental conditions, leading to reduced stability and bioactivity during processing and digestion.

Scope: To overcome the stability challenges, double emulsions (W1/O/W2) have emerged as an innovative encapsulation and delivery system for bioactive compounds. This review explores the structural characteristics of double emulsions, their encapsulation mechanisms, and models describing release kinetics under digestive conditions. Factors influencing emulsion stability—including emulsifier type, droplet size, phase ratio, and processing conditions—are critically discussed. Studies demonstrate that double emulsions can significantly enhance the stability, bioaccessibility, and controlled release of sensitive compounds such as betalains, while maintaining desirable sensory and functional properties. Their successful incorporation into various food matrices, including beverages, dairy, and bakery products, has resulted in improved nutritional profiles, antioxidant retention, and extended shelf life. Additionally, industrial scalability and formulation challenges are examined, emphasizing the importance of optimizing processing parameters and ensuring consumer acceptability.

Conclusion: Double emulsions provide a novel approach to improve the stability and delivery of bioactive ingredients in functional foods. Future research and development should focus on optimizing their design and functionality, enabling widespread application across the food industry and healthier food products.

Link: https://doi.org/10.1186/s43088-025-00711-0

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

An Experimental Investigation Of The Partial Substitution Of Steel Slag As Fine Aggregate In Concrete

AUTHOR :

Shinde, S.; P.; R.; Sathe, S.; Ingle, G.

JOURNAL NAME:

ASEAN Engineering Journal (Vol.-15, Issue-4)

DETAILS:

Published on December 2025

ABSTRACT:

The use of steel slag as aggregate in concrete is one environmentally friendly method to reduce the detrimental consequences of the depletion of natural mineral resources. The behavior of steel slag sand concrete (SSC) under compression is examined in this paper. A set of cube tests under compression are reported, with the results evaluated. The experiments used SSCs integrated with steel slag at volume substitutions of 0, 10, 20, 30, and 40% for fine aggregate. The findings indicate that concrete's compressive strengths can be raised by including steel slag as fine aggregate. The strength of compression of SSC first rises as the amount of steel slag increases when loading is applied. Compressive strength falls once the ideal percentage of steel slag is added. Water permeability, acid attack, and rapid chloride penetration test are conducted for the durability test. The addition of steel slag to concrete increases the material's ductility, durability, and resistance to cracking. To achieve enhanced performance for SSC under compression, 20% of steel slag should be used as fine aggregate.

Link: https://doi.org/10.11113/aej.v15.23293

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Designing AI-Enabled Engineering Courses: E-AIP Framework for Learning Outcomes, Process Evidence, and Integrity

AUTHOR :

Khataan, A.M.; Bapat, G.; Rahaman, S.; Kumar, A.

JOURNAL NAME:

International Journal of Engineering Pedagogy (Vol.-15, Issue-7)

DETAILS:

Published on November 2025

ABSTRACT:

Artificial intelligence (AI) is entering engineering courses rapidly, yet tool-led adoption can weaken assessment validity and academic integrity. This paper presents the E-AIP (Engineering–AI Pedagogy) Framework, which centers three pillars learning Outcomes, Process Evidence, and Integrity & Ethics Guardrails and links them to design levers (AI function, task authenticity, feedback granularity, locus of agency). We define seven constructs, state eight propositions about alignment, validity moderation, authenticity, and agency, and operationalize E-AIP through a compact matrix (AI function × outcome type with required process evidence and guardrails). Two design patterns (CS1 debug-with-defense; circuits param-twins) illustrate classroom use; a lightweight adoption toolkit (two rubrics and an integrity or privacy checklist) supports immediate deployment. Additional patterns and full matrices appear in the online supplement. E-AIP enables instructors to capture AI's benefits while preserving what scores validly claim to measure.

Link: https://doi.org/10.3991/ijep.v15i7.59045

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Transforming Engineering Pedagogy through the Lens of the Sustainable Development Goals

AUTHOR :

Kaur, S.; Devi, S.D.; Banerjee, S.

JOURNAL NAME:

International Journal of Engineering Pedagogy (Vol.-15, Issue-7)

DETAILS:

Published on November 2025

ABSTRACT:

Transforming engineering pedagogy is imperative in an era defined by global sustainability challenges. The United Nations' Sustainable Development Goals (SDGs) offer innovative teaching methodologies by embedding necessary competencies to tackle complex virtual and real-world problems. The paper integrates engineering courses with SDG-6, SDG-7, SDG-9, and SDG-11. By viewing education pedagogy through the lens of the SDGs, this paper positions the shift from the traditional content-based model towards project- and problem-based models that cultivate ethical solutions, critical thinking, interdisciplinary collaboration, and the transformation of scholarly, industrial, and societal perspectives. At a policy level, this transformation entails reforming the curriculum and assessment framework that values both societal impact and technical competence. By viewing engineering education through the lens of the SDGs, this study positions pedagogy as a catalyst for preparing future engineers not only as innovators but also as responsible agents of sustainable global development.

Link: https://doi.org/10.3991/ijep.v15i7.59049

TITLE:

Description of Protosticta sooryaprakashi sp. nov. (Odonata: Zygoptera: Platystictidae) from the Western Ghats, India

AUTHOR :

K.A.R.M.; Chandran, A.V.; Sawant, D.; Y.P.M.; Koparde, P.; Kunte, K.

JOURNAL NAME:

Zootaxa (Vol.- 5723, Issue-3)

DETAILS:

Published on November 2025

ABSTRACT:

We erect a new species of damselfly, Protosticta sooryaprakashi sp. nov. from Sampaje, Kodagu District, Karnataka, India (type specimens are deposited in the Biodiversity Lab Research Collections, currently held at the National Centre for Biological Sciences, Bengaluru). Morphologically, this species shows close affinity with the members of the P. sanguinostigma Fraser, 1922 group; however, it is distinguished by unique features in its prothorax, caudal appendages, and accessory genitalia. Additionally, the analyses also revealed a significant genetic distance between the new species and other species of this group. We present an updated identification key to the males of Protosticta Selys, 1885 species in the Western Ghats, incorporating newly available diagnostic information.

Link: https://doi.org/10.11646/zootaxa.5723.3.3

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Review on Potential Application of Graphitic Carbon Nitride as Adsorbent/Photocatalyst for Wastewater Treatment

AUTHOR :

Polshettiwar, P.; Topare, N.S.; Khedkar, S.; Thorat, P.; Chavali, M.; Khan, A.; Choudhari, V.; Satish, S.

JOURNAL NAME:

AIP Conference Proceedings (vol.-3360, Issue-1)

DETAILS:

Published on 11 November 2025

ABSTRACT:

Interest in using graphitic carbon nitride (GCN) to treat water and wastewater has grown significantly. The intriguing, layered structure, low-cost and rapid production, and GCN photocatalytic activity have attracted much attention. One promising and long-term photocatalyst that could revolutionize our approach to energy and environmental crises is GCN. An overview of the process for producing and characterizing GCN is provided in this review, along with examples of its use in dyes and heavy metal treatment. Also mentioned is a photocatalyst/adsorbent based on GCN that can remove pollutants by photocatalytic degradation. In addition, this review offers a few viewpoints, discusses various limitations, and obstacles in the exploration of GCN.

Link: https://doi.org/10.1063/5.0305973

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Influence of Nano-Catalysts on Biodiesel Synthesis: A Comprehensive Review

AUTHOR :

Kale, A.S.; Topare, N.S.; Khedkar, S.; Thorat, P.; Khan, A.; Patil-Shinde, V.; Chavali, M.

JOURNAL NAME:

AIP Conference Proceedings (vol.-3360, Issue-1)

DETAILS:

Published on 11 November 2025

ABSTRACT:

Given the growing depletion of traditional fossil fuels, finding new, sustainable energy sources is essential. Maybe biodiesel will become the better option and displace traditional fuels. Catalysts that are both inexpensive and safe for the environment will probably be utilized to make biodiesel by transesterification of a variety of feedstocks. Many studies have examined homogeneous/heterogeneous catalysis for oil-to-biodiesel conversion, and new heterogeneous catalysts are being studied, and new heterogeneous catalysts are now being investigated. Because of their high efficiency, researchers are becoming more interested in employing nano-catalysts to produce biodiesel. The potential of nanocatalysts to increase the sustainability and efficiency of biodiesel synthesis from a range of feedstocks is evaluated in this analysis. It highlights the financial and ecological benefits of using nanocatalysts, which circumvent competing with agricultural resources while overcoming challenges including high costs, low conversion rates, and intricate purifying procedures. The review discusses the functions of carbon-based nanocatalysts, metal oxides, and magnetic nanoparticles in the generation of biodiesel. When used in traditional biodiesel synthesis, nanocatalysts can increase productivity, lessen their negative effects on the environment, and save production expenses. An interest in the development and current uses of nanoparticles in biodiesel synthesis and property enhancement has been sparked by this review. It also provides a framework for individuals who are just starting in this area.

Link: https://doi.org/10.1063/5.0305974

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Ultrasonic and Mechanical Stirring Methods for Biodiesel Production Using Cottonseed Oil (CSO) and Sesame Oil (SO)

AUTHOR :

Topare, N.S.; Khedkar, S.; Thorat, P.; Paturi, U.M.R.; Khan, A.; Chavali, M.; Faheem, S.M.

JOURNAL NAME:

AIP Conference Proceedings (vol.-3360, Issue-1)

DETAILS:

Published on 11 November 2025

ABSTRACT:

A catalyst converts vegetable oils to esters at a lower reaction temperature and faster than fatty acids, which can be transesterified with methanol to make biodiesel. Homogenous and heterogeneous catalysts have been identified, and many researchers prefer homogenous catalysts since they produce biodiesel faster. Homogeneous catalysts are popular in biodiesel synthesis due to their repeatability and simplicity of separation. Biodiesel production using low-frequency ultrasonic radiation at 20 kHz and a mechanical stirrer is detailed. This project will use cottonseed oil (CSO) and sesame oil (SO). Experimental conditions included a 1:5 oil-to-alcohol molar ratio, a 60-minute reaction period, and a 68°C-reaction temperature. As catalysts, sodium methoxide (CH3NaO) and potassium carbonate (K2CO3) were utilized at 0.25, 0.50, and 0.75 wt.%. The chemical and physical properties of cottonseed oil (CSO) and sesame oil (SO) biodiesel have been identified. The ultrasound biodiesel synthesis technology is viable, much simpler and more effective than mechanical stirring.

Link: https://doi.org/10.1063/5.0305977

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Review on Adsorption: An Eco-Friendly Method for Dye Removal

AUTHOR :

Topare, N.S.; Raut-Jadhav, S.; Khedkar, S.; Thorat, P.

JOURNAL NAME:

AIP Conference Proceedings (vol.-3360, Issue-1)

DETAILS:

Published on 11 November 2025

ABSTRACT:

The planet is facing significant threats of air, soil, and water contamination as a result of various human activities. Environmental impacts in general have become much more severe over the years, but water pollution is quite advanced. Organic pollutants can be found in a variety of industrial wastewater sources, including textile waste liquids. Direct discharge into natural waters would pose a significant risk to the marine environment. Although direct discharge to the wastewater system will adversely affect the future treatment of biological wastewater. Adsorption is low cost, has a high binding capacity, and the primary role of microorganisms drew the scientific community's attention as an alternative method for removal. Also, adsorption is a promising alternative for replacing or supplementing the current wastewater treatment processes. This approach chooses the optimal mode and adsorbent based on simplicity, nontoxicity, low cost, and mild circumstances. This review article discusses dye adsorption by a variety of adsorbents, as well as the various physicochemical factors and dye adsorption mechanisms.

Link: https://doi.org/10.1063/5.0305975

Since 1983

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TITLE:

AI-driven development: Innovating biomedical polymer materials

AUTHOR :

Kalbhor, I.; Chakote, S.; Dabhade, R.; Sharma, P.; Gaikwad, S.; Agarwal, R.; Satish, S.

JOURNAL NAME:

In book: Artificial Intelligence in Polymer Science and Nanotechnology

DETAILS:

Published on June 2025

ABSTRACT:

In the era of artificial intelligence (AIartificial intelligence), polymeric biomaterialsbiomaterials alter efficiency and precision in material design and propertiesproperties. AI-driven modelsAI-driven models are leveraged for predicting, optimizing and developing novel biomaterials. A data-driven model for enhancing material propertiesmaterial properties may improve the development of polymeric biomaterialspolymeric biomaterials. This chapter depicts various AIartificial intelligence-driven models, techniques, their advantagesadvantages, and disadvantages for a robust and adequate AIartificial intelligence model for innovating polymeric biomaterials. Advanced computational tools with AIartificial intelligence-driven models for novel polymeric biomaterials may help researchers understand and redefine the landscape of biomedical polymerspolymers for innovating, patient-centric solutions in modern medicine.

Link: https://doi.org/10.1515/9783111681870-008

TITLE:

An Autonomous Mobile Robot Simulation Using Navigation (NAV2) and Simultaneous Localization and Mapping (SLAM) in ROS2

AUTHOR :

Patil, R.V.

JOURNAL NAME:

Transforming Industries
Capturing the Potential of IIoT and ML in the Era of Industry 5.0(Book Chapter)

DETAILS:

Published on October 2025

ABSTRACT:

This chapter proposes a method for simulating a mobile robot in ROS2 by combining the NAV2 and SLAM frameworks. The main objective is to simulate and test a mobile robot in a controlled simulated environment. The robot model and its interactions with surrounding obstacles are investigated in the study using a simulated environment constructed with Gazebo. Real-time data collection is incorporated into the robot model using depth cameras and LiDAR; this data is examined and presented in RVIZ. The outcomes demonstrate that path planning and mapping are successful, enabling the robot to navigate on its own while dodging hazards. It demonstrates how well NAV2 and SLAM complement one another to enhance the autonomy and usefulness of autonomous mobile robots (AMRs) in their natural environments.

Link: https://doi.org/10.1201/9781003538271-14

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

RFID Integrated with Human Interface System for Autonomous Shopping Tram with Integrated Bill Generator

AUTHOR :

Patil, R.V.

JOURNAL NAME:

Transforming Industries: Capturing the Potential of IIoT and ML in the Era of Industry 5.0 (Book Chapter)

DETAILS:

Published on October 2025

ABSTRACT:

An important step forward in automated tram systems for retail automation has been made with the use of RFID technology to create an integrated bill generator in the system. The design of an automatic shopping tram with an integrated bill generator is presented. Radio frequency identification (RFID) technology will be used to accomplish the automatic bill generator, while ultrasonic and infrared (IR) sensors are integrated into the automated system to enable autonomous movement. LCD_I2C will be installed in the tram to improve user convenience and the shopping experience. The Arduino Uno R3 microcontroller will be used to integrate all the components. By combining RFID modules specifically, the RC522, an Arduino Uno microcontroller, LCD, and ultrasonic and infrared sensors, the system offers users a user-friendly interface and satisfied customers.

Link: https://doi.org/10.1201/9781003538271-25

Since 1983

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Floral traits, pollination syndromes, and nectar resources in tropical plants of Western Ghats

AUTHOR :

Patwardhan, A.; Tadwalkar, M.; Joglekar, A.; Sonne, M.; Pawar, V.; Mestry, P.; Kulkarni, S.; Kashikar, A.; Pachpor, T.

JOURNAL NAME:

Journal of Threatened Taxa (Vol.-17, Issue-10)

DETAILS:

Published on 26 October 2025

ABSTRACT:

Tropical regions are known to have a high percentage of animal-pollinated plants. This study explores the natural history of pollination in an understudied biodiversity hotspot, the tropical forests of India's Western Ghats. It is the first-ever attempt to gain insights into three critical aspects of pollination simultaneously, i.e., pollination syndromes, floral visitors, and standing nectar crop. Data on the attributes of floral visitors of 62 plant species were collected through regular field visits for three years allowing for sampling across seasons. 'Tube' was the most dominant flower type (20) followed by 'Dish to bowl' with 18 species, 'Brush or Head' (13), and 'Gullet' with nine species. The range of nectar quantity per flower varied from 0.05–13.7 µL. Nearly 40 percent of plant species observed by us have only Lepidopteran visitors. Fifteen plant species were visited by hymenopterans and lepidopterans, whereas five plant species had hymenopteran visitors only. In the light of rapidly declining pollinator diversity, our study highlights the significance of floral visitors in the pollination of some conservation-significant species, as well as points to determinants of floral visitation and success.

Link: https://doi.org/10.11609/jott.9755.17.10.27637-27650

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Balancing Automation and Human Touch: Strategies for Nurturing the Human Element in an AI-augmented Workplace

AUTHOR :

Singh, M.; Srivastava, S.; Paigude, S.; Srivastava, S.P.

JOURNAL NAME:

In book: Impact of Artificial Intelligence on Data-Driven Decision Making in HR for Revolutionizing Organizational Growth

DETAILS:

Published on October 2025

ABSTRACT:

In this era of transition from traditional, people-centric Human Resources (HR) to data-driven and unbiased approaches, this collection not only highlights the revolutionary potential of Artificial Intelligence (AI) but also addresses the challenges and ethical considerations of AI deployment in the workplace. Impact of Artificial Intelligence on Data-Driven Decision Making in HR for Revolutionizing Organizational Growth explores the transformative role of artificial intelligence in reshaping HR, recruitment, and organisational culture. As AI becomes increasingly embedded in organisational systems, it offers unprecedented opportunities to enhance HR operations–from advanced analytics that predict employee behaviours and trends, to optimising structures that foster innovation and growth. By providing a comprehensive overview of AI's impact on HR, this insightful volume offers HR leaders, C-suite executives, organisational strategists, and academics the tools to harness data-driven decision-making for organisational growth in a rapidly changing world.

Link: https://doi.org/10.1108/978-1-83662-982-520251008

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Ecological Impact of Chemical Pollution on Herbs of Medicinal Value

AUTHOR :

Kothawade, S.; Patil, P.; Bole, S.; Suryawanshi, J.; Raut, K.; Wagh, V.

JOURNAL NAME:

In book: Ecotoxicity and Herbal Health

DETAILS:

Published on August 2025

ABSTRACT:

This chapter explores the complex relationship between chemical pollution and the ecological health of medicinal herbs, including the effects on ecosystem services, biodiversity, and plant-animal interactions. An overview of the significance of medicinal herbs is given in the introduction, with a focus on the vital importance of ecological balance. The main area of research concern is chemical pollution and how it affects these priceless herbs.

The chapter covers historical viewpoints on the use of medicinal herbs, ecological roles, types, and sources of chemical pollution, as well as earlier research that highlights knowledge gaps. The emphasis switches to a few chosen medicinal herbs, explaining their chemical makeup, ecological significance, and pollution susceptibility.

The chapter examines the sources, pathways of exposure, and mechanisms of uptake of chemicals, as well as the physiological and biochemical reactions of medicinal herbs. The effects of biodiversity on ecosystems include modifications to plant-animal interactions, adjustments to population dynamics, and effects on related fauna. Changes in community structure, soil and water quality, and population dynamics are all assessed in the assessment section. Policy recommendations for pollution control are identified, along with mitigation strategies and restoration techniques.

The chapter ends with recommendations for future research, highlighting areas that need more study, new risks to medicinal herbs, and how to incorporate new technologies into ecological impact assessments. This thorough investigation advances knowledge of the complex interactions between chemical pollution and the ecological resilience of medicinal herbs.

Link: https://doi.org/10.1201/9781779640437-3

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Smart Door Locking System for Children Using HC-SR04 and IoT Technology

AUTHOR :

Melinda, M.; Yunidar, Y.; Khairia, S.; Miftahujjannah, R.; Sakarkar, G.; Basir, N.

JOURNAL NAME:

Jurnal Nasional Teknik Elektro

DETAILS:

Published on 31 July 2025

ABSTRACT:

The increasing incidence of minors accessing hazardous indoor areas—such as staircases, balconies, and rooms with sharp objects—raises serious safety concerns, often due to insufficient parental supervision. This study proposes an Internet of Things (IoT)-based automatic door lock system to enhance child safety in home environments. The system integrates dual ultrasonic sensors for distance and height detection, a KY-037 sound sensor, and an ESP32-CAM for real-time video monitoring, all accessible via a web interface. A key novelty lies in the integration of multi-sensor spatial awareness with live surveillance, enabling automated control and proactive safety features. Tested on ten children aged 4 to 6 years, the system achieved a 90% success rate in locking the door when a child under 120 cm approached within 1 meter, with an average response time of approximately 2 seconds. A sound-based alarm is also triggered when noise levels exceed 120 decibels, serving as an emergency alert. However, a 10% false negative rate was observed when children were detected at distances of 1.3 to 1.5 meters, suggesting the need for further sensor calibration. Overall, the system demonstrates strong potential as a scalable and cost-effective smart home safety solution, combining automation, real-time monitoring, and child-specific access control. Future work should focus on improving detection accuracy and extending functionality for multi-object scenarios.

Link: https://doi.org/10.25077/jnte.v14n2.1293.2025

TITLE:

Design and Development of a Smokeless Stove for Environmental Protection of Smart City

AUTHOR :

Nayak, R.C.; Roul, M.K.; Kulkarni, M.V.; Roul, P.D.; Vijaybhai, S.B.

JOURNAL NAME:

Indian Journal of Environmental Protection (Vol.-45, Issue-7)

DETAILS:

Published on July 2025

ABSTRACT:

In rural Odisha, India, traditional biomass cook stoves, or chulas, are commonly used for cooking three times a day. These stoves burn wood, paddy husks and other agricultural residues, which results in substantial indoor air pollution due to inefficient combustion. The smoke produced contains harmful pollutants, including particulate matter (PM10, PM2.5), carbon monoxide (CO), nitrogen oxides (NOx), sulphur oxides (SOx), polycyclic aromatic hydrocarbons (PAHs), formaldehyde and polychlorinated dibenzofurans (PCDFs). Exposure to these pollutants has been linked to various adverse health effects, such as chronic respiratory diseases, cardiovascular issues, skin irritations and exacerbation of pre-existing conditions. Additionally, the poor air quality from these stoves contributes to broader environmental concerns, including outdoor air pollution. To address these problems, a new smokeless chula has been developed, incorporating a dual-fuel system that combines primary fuel as biomass (such as wood and agricultural waste) with secondary fuel as superheated steam. This design aims to improve combustion efficiency and reduce harmful emissions. Utilizing both primary and secondary fuels, stove significantly improves thermal efficiency while reducing primary fuel consumption. Its environmentally-friendly design guarantees negligible smoke and achieves complete combustion. This work presents a stove design incorporating modern techniques to decrease emissions and improve thermal efficiency. The stove minimizes environmental damage and improves public health by concentrating on these characteristics. Compared to traditional stoves, this model establishes a 50% increase in efficiency due to forced draught and superheated steam. It also decreases cooking time, fuel usage and CO emissions.

Link: https://www.e-ijep.co.in/45-7-670-676/

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Anticholinergic and sedative medication burden in older adults: an observational analysis

AUTHOR :

Syed, J.; Pereira, P.; Tejaswini, C.J.; Avarebeel, S.; Ramesh, K.; Ramesh, M.; Undela, K.; Chalasani, S.H.

JOURNAL NAME:

Geriatric Care (Vol.-11, Issue-2)

DETAILS:

Published on November 2025

ABSTRACT:

The cumulative effects of multiple medications with anticholinergic and sedative properties, referred to as anticholinergic and sedative burdens, can result in adverse effects in older adults. This study aimed to evaluate the exposure of older adult patients to anticholinergic and sedative medications. This cross-sectional study included older adult patients (aged ≥60 years) admitted to the geriatric inpatient wards. The anticholinergic burden was assessed using the CRIDECO scale, and the Drug Burden Index (DBI) quantified the cumulative anticholinergic and sedative burden. The data obtained were categorically assessed and are represented as [n (%)]. The associations between medication burden and health outcomes were analyzed using appropriate tests. The mean age of the participants in the study cohort was 72.80±7.41 years. Among the participants, 29.17% were administered at least one anticholinergic medication, with a mean cumulative Anticholinergic Cognitive Burden (ACB) score of 2.49±2.00. The majority (94.34%) had a cumulative DBI score of ≥1, with a mean DBI score of 3.14±1.55. Linear regression analysis demonstrated that both ACB and DBI were significant predictors of the length of hospitalization (ACB: p=0.002; DBI: p=0.001), whereas only DBI was associated with the Charlson Comorbidity Index (p=0.001). Logistic regression revealed that ACB scores significantly predicted cognitive decline (p=0.047). This study highlights the high prevalence of anticholinergic and sedative medication use among hospitalized older adult patients and its association with adverse health outcomes. Regular medication reviews and targeted deprescribing strategies are essential to reduce the medication burden in this vulnerable population.

Link: https://doi.org/10.3390/pharmacy13050144

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Innovative Probe Drilling Technique for Ground Condition Evaluation in the Tunnelling of Lesser Himalaya – A Case study

AUTHOR :

Kumar, M.; Potnis, S.; R.

JOURNAL NAME:

Water and Energy International (Vol.-68r, Issue-3)

DETAILS:

Published on 2025

ABSTRACT:

Geological difficulties specific to the Himalayas include a variety of rock kinds, seismic activity, and shifting weathering patterns. Conventional exploration techniques frequently fail to record the minute data required for precise tunnel design and construction scheduling. This paper presents an entirely novel form of probe drilling specifically designed to evaluate ground conditions in the Himalayan region. It focuses on using this technique for tunnel building endeavours for New BG line of Rishikesh-Karanprayag project. Modern drilling technologies are combined with advanced geophysical and geotechnical analysis in the suggested probe drilling approach. The technique enables continuous monitoring of subsurface conditions and real-time data capture through the use of specialized drilling instruments and sensors. In order to overcome the drawbacks of traditional approaches and provide a more accurate picture of the intricate geological layers that are common in the Himalayan region, the study highlights the significance of in-situ data collection. The practicality and dependability of the probe drilling method are illustrated via case study and field tests. The outcomes demonstrate the technique's capacity to offer comprehensive details on structural integrity, possible dangers, and rock properties—all of which are essential for determining if tunnel construction is economically feasible. The approach is also a viable option for infrastructure projects in difficult terrain because to its cost-effectiveness and flexibility to different geological formations. The results of this study provide a dependable and effective tool for planning tunnel building, furthering site investigation procedures in the Himalayas. The suggested probe drilling method could improve geological evaluations' accuracy, resulting in safer and more durable tunnel infrastructure in the Himalayan region and elsewhere. In order to meet the particular geological complexity of mountainous terrains, this study promotes the implementation of novel exploration techniques, supporting strong and sustainable engineering practices in tunnel construction projects.

.Link:
https://www.scopus.com/pages/publications/105024762518

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Defect detection-based sewage overflow forecasting using CDS2KNN and ESI-aroma

AUTHOR :

Gawali, V.S.

JOURNAL NAME:

International Journal of Mining and Mineral Engineering (Vol.-16, Issue-4)

DETAILS:

Published on 23 December 2025

ABSTRACT:

Global changes in climatic conditions and urbanised environments require reliable drainage systems (DSs). By analysing the characteristics of DS, the sewage flood in urban areas is managed with prior forecasting. However, prevailing works failed to forecast the future overflow of sewage due to their defects. This paper forecasts future overflow based on defects and water level in the drainage. The process starts with data collection. Then, the data undergoes preprocessing. Further, the dimensionality of the data is reduced with MDS. Water level, gas leakage, and defects in drainage are categorised utilising CDS2KNN. The images for defect detection are then collected and pre-processed. The resultant image is segmented utilising PPSOA. Thereafter, defect types are categorised utilising CDS2KNN. Seasonal data is collected and clustered via the KGHWMA model. The future overflow is forecasted utilising ESI-AROMA. When compared with the prevailing methods, the proposed system attains an improved prediction accuracy (98.987%) and root mean square error (RMSE) (0.12).

Link: https://doi.org/10.1504/IJMME.2025.150836

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Development of Biochar-Based Sustainable Corrosion-Resistant Coating †

AUTHOR :

Zade, G.; Kulkarni, M.

JOURNAL NAME:

Engineering Proceedings (Vol.-105, Issue-1)

DETAILS:

Published on August 2025

ABSTRACT:

Conventional protective coatings based on petroleum raw materials have certain limitations in terms of their availability, environmental pollution, and sustainability. Therefore, this research successfully investigates the potential of sheep wool-derived biochar to develop a sustainable, high-performance protective coating. Two variants of biochar, namely SW800 and SW1000, were developed by pyrolyzing sheep wool at 800 °C and at 1000 °C for 1 h, respectively. The prepared samples were characterized using FTIR, FESEM-EDX, and XRD analyses to confirm the structural and elemental differences between both biochar samples. Furthermore, biochar-based epoxy coatings were developed by varying the concentration of prepared biochar from 1% to 5%. The coating performance was evaluated for its aesthetic, mechanical, chemical resistance, and hydrophobicity. Crucially, this study demonstrated that biochar inclusion did not compromise critical mechanical and chemical properties like adhesion (5B), flexibility (7 mm), scratch hardness (3500 gms), pencil hardness (3H), acid-alkali resistance, and solvent rub test (rating 5). However, a key finding of this research is that the incorporation of biochar into an epoxy coating resulted in a significant improvement in hydrophobicity, which is measured using water contact angle. The incorporation of SW800 and SW1000 into coating formulations at varying concentrations resulted in an increase in water angle of approximately 18% and 20%, respectively. The outcomes of this project establish biochar-based coatings as a promising solution for eco-friendly and high-performance protective applications.

Link: https://doi.org/10.3390/engproc2025105005

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Mapping CSR and employee organizational commitment: A bibliometric review of thematic evolution and research frontiers

AUTHOR :

Gadekar, M.; Sharma, E.

JOURNAL NAME:

Problems and Perspectives in Management (Vol.-23, Issue-4)

DETAILS:

Published on 18 December 2025

ABSTRACT:

Amidst growing employee-related challenges such as turnover, declining trust, and disengagement, examining the impact of CSR initiatives on behavioral outcomes, such as organizational commitment, becomes critical. This study systematically maps the social, conceptual, and intellectual structure of fragmented research on the CSR–organizational commitment link. Using Biblioshiny and VOS viewer, it analyzes 432 documents from the Web of Science spanning 2005–2025. Results show a consistent growth in publications over two decades, with China as the most prolific contributor and the UK as the top international collaborator. The study showcases recurring contributions from multiple journals and authors. Keyword co-occurrence reveals eight thematic clusters, and thematic evolution shows how the CSR–employee commitment relationship has expanded to include HRM practices, corporate governance, and the underexplored SME context. Analysis of trend topics indicates a clear shift in research focus from ethics and stakeholder theory to literature examining intervening variables in the CSR–organizational commitment association. The study suggests future research directions, including investigating new mediator–moderator variables, deeper integration of CSR and HRM, and greater focus on SMEs, where empirical data are limited. These findings reflect the growing interdisciplinary nature of CSR–employee commitment research, which merges sustainability, behavioral, and organizational perspectives.

Link: https://doi.org/10.21511/ppm.23(4).2025.39

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Curved Parasitic Element-Based Quad-element Antenna for High-Gain Millimeter Wave 5G Communications

AUTHOR :

Dabhade, M.K.; Warhade, K.K.

JOURNAL NAME:

Progress in Electromagnetics Research C (Vol.-162)

DETAILS:

Published on December 2025

ABSTRACT:

This paper proposes a novel four-port MIMO antenna specifically designed for millimeter-wave (mm-wave) 5G applications. The antenna features a compact symmetric layout measuring 22 mm × 22 mm, corresponding to approximately 2.7λ × 2.7λ at 37 GHz. The prototype is fabricated on a Rogers RT Duroid 5880 substrate ($\varepsilon r = 2.2$, $\tan\delta = 0.0009$, $h = 0.8$ mm) to ensure low loss and stable performance at high frequencies. The antenna operates effectively over two targeted frequency bands, 37-41 GHz and 42-43.5 GHz, making it suitable for high-data-rate, short-range communication systems in emerging 5G networks. The structure is evolved through multiple design stages using strategically placed curved parasitic elements to achieve dual-band operation, high isolation, and enhanced gain. Experimental validation using a vector network analyzer and anechoic chamber confirms good agreement between simulated and measured S-parameters, with isolation better than -20 dB. The antenna demonstrates a measured gain between 9.3 and 9.7 dBi, with simulated peaks up to 11 dBi. Far-field pattern measurements exhibit stable bidirectional radiation with low cross-polarization and well-defined main lobes at both 38 GHz and 42 GHz. MIMO performance metrics such as ECC < 0.01, DG ≈ 10 dB, MEG ≈ -3 dB, and CCL < 0.4 bps/Hz confirm efficient multi-port operation. The proposed antenna thus offers a compact, high-isolation, high-gain solution for next-generation mm-wave 5G MIMO systems.

Link: https://doi.org/10.2528/PIERC25091803

TITLE:

Unsupervised Machine Learning for Bowler Performance Analysis in Indian Premier League (IPL) and Cross-League Prediction

AUTHOR :

V.K.; Metkewar, P.S.; Deshpande, D.S.; Bokhare, A.; Dewi, D.A.; F.

JOURNAL NAME:

Journal of Advances in Information Technology (vol.-16, Issue-12)

DETAILS:

Published on December 2025

ABSTRACT:

The selection of players in cricket, particularly in formats like the Indian Premier League (IPL), is crucial for a team's success. This process involves analyzing various player attributes, including their batting and bowling performances, to create a balanced team. In cricket, each player's role—batsman, bowler, or all-rounder—must be clearly defined. This clarity helps in optimizing team dynamics and ensuring that players can perform at their best. For instance, a well-rounded team might include specialists who excel in specific areas, during matches. Recent advancements in Artificial Intelligence (AI) have opened new avenues for player analysis. Unassisted AI models can evaluate bowlers based on their historical performances in the IPL. By leveraging data from past matches, these models can predict future performances, aiding selectors in making informed decisions about player selection. This study employs predictive modelling to analyze bowler performance using unsupervised learning. The developed AI model can extend its applicability beyond the IPL, offering insights into bowlers from other leagues. There are some discriminative features of players' performance as well as team strategy, which can be obtained by applying machine learning in Bowler Performance Analysis. This data-driven approach provides a decision-making and fast response possibility to managers and analysts in playing T20 matches 2018. By combining various techniques, e.g., K-means and mean-shift, teams are also equipped to select players more effectively, establish strategic and success rate, etc.

Link: https://doi.org/10.12720/jait.16.12.1675-1684

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

A Comprehensive Review of Optimal Path Planning Techniques for Industrial Robots

AUTHOR :

Y.; B.K.; Bhojane, P.K.

JOURNAL NAME:

In book: Metaheuristics in Engineering Applications

DETAILS:

Published on November 2025

ABSTRACT:

This chapter presents an in-depth study of optimal path-planning algorithms for industrial robots, highlighting their evolution, key strategies, and ongoing research. It explores sampling-based algorithms (RRT, PRM), evolutionary algorithms (GA, PSO, WOA, ACO, DE), graph search algorithms (A*, D*), interpolation methods (S-curve, Cubic B-Spline), and machine learning approaches (Neural Networks, Fuzzy Logic, Deep Reinforcement Learning). These algorithms are used to achieve efficient, collision-free path planning in robotic systems. The review begins with foundational principles of each method and examines their effectiveness in collision avoidance. It evaluates how robots adapt paths in static and dynamic environments using sensors and adaptive control to complete tasks under varying conditions. A comparative analysis underscores the strengths and weaknesses of each approach. Notably, limited research exists on path planning in dynamic environments, especially involving multiple robotic arms. Among available methods, evolutionary algorithms like PSO and GA—and their hybrids—are particularly effective due to their adaptability to changing conditions. Overall, this review provides a comprehensive overview of algorithms essential for integrating robotic automation into modern manufacturing, offering valuable insights for future developments in optimal path planning.

Link: https://doi.org/10.1201/9781003561484-1

TITLE:

January 2026 issue first authors

AUTHOR :

Shrivastav, P.; Fois, M.; Ibrahim, M.; Zhang, W.; An, J.; Kobyra, J.A.; Li, Y.-R.

JOURNAL NAME:

Trends in Pharmacological Sciences

DETAILS:

Published on 23 December 2025

ABSTRACT:

Meet the first authors' is a TrendsTalk series launched in Trends in Pharmacological Sciences in 2026 by Dr Jerry Madukwe, Editor-in-Chief of the journal. These articles feature the first authors of review and opinion pieces published in each monthly issue. The series offers a platform for authors to share their journey to becoming scientists, their current roles and sources of inspiration, personal perspectives on current challenges in their fields, and their thoughts on future research directions. It also highlights opportunities for collaborations, new ventures they are eager to pursue, and advice for young scientists.

Link: https://doi.org/10.1016/j.tips.2025.12.002

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Balancing style, function and sustainability: a framework for textile product design decisions

AUTHOR :

G.; Ghag, N.

.

JOURNAL NAME:

Research Journal of Textile and Apparel

DETAILS:

Published on 17 December 2025

ABSTRACT:

Purpose: In today's constantly evolving textile and fashion industry, product design decision-making must involve multiple considerations, including aesthetics, functionality and sustainability. As consumer tastes evolve and environmental concerns rise, determining the most suitable design choices is an umbrella process with numerous interrelated factors. The research aims to examine and score criteria for textile product design by structuring them into three broad dimensions: aesthetic, functional and sustainable.

Design/methodology/approach: A systematic review of literature was followed to enumerate the distinguishing criteria for every dimension, along with expert consultations in context for validity. Their evaluation was carried out with a formal decision support model, best worst method, which strikes an optimal balance between data economy and comparison consistency.

Findings: Functional aspects such as comfort and stamina have the highest weightage, followed by sustainability factors such as use of green materials and recyclability. Aesthetic factors, being as subjective in nature as they are, also emerged to be important, particularly in highly engaging products.

Practical implications: The findings provide theoretical implications by extending current design evaluation models with an encompassing, integrated framework. Managerially, the framework offers designers, product developers and sustainability officers a decision-making tool to map product innovations to market and environmental standards.

Originality/value: The study recognizes the strengths of systematic, criterion-based evaluation of improving competitiveness and customer satisfaction in products. The drawbacks are generalizability and expert judgment subjectivity, but follow-up studies can be executed using real-time consumer knowledge and design adaptation feedback loops. The study helps to bridge the knowledge gap between sustainable innovation and customer-centric design for the textile sector.

Link: https://doi.org/10.1108/RJTA-05-2025-0109

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Influence of leader humility on subordinates' knowledge-sharing: exploring the boundary conditions

AUTHOR :

Upadhyay, S.; Singh, P.; Ghosh, P.; Sarkar, S.

JOURNAL NAME:

Journal of Knowledge Management

DETAILS:

Published on 18 December 2025

ABSTRACT:

Purpose: Among the human aspects of leaders, humility has gained traction in recent explorations. Subordinates' psychological safety has been established to mediate between leader humility and subordinates' knowledge sharing. However, because knowledge sharing is a complicated phenomenon driven by situational demands and individual characteristics, the authors project the presence of boundary conditions in the influence of leader humility on knowledge sharing through psychological safety. The purpose of this study is to investigate the moderating effect of leader competence on the relationship between leader humility and psychological safety, as well as the moderating effect of subordinates' occupational self-efficacy (OSE) on the relationship between psychological safety and subordinates' knowledge sharing, within the framework of the social information processing (SIP) theory.

Design/methodology/approach: The authors used a time-lagged, three-wave survey-based questionnaire. Responses from 392 employees working in the Indian IT organizations, collected by using judgmental sampling techniques, were analyzed using partial least squares structural equation modeling.

Findings: Results confirm that leader humility influences subordinates' knowledge sharing through psychological safety. Findings also prove moderation by leader competence between leader humility and psychological safety and by OSE between psychological safety and subordinates' knowledge sharing.

Originality/value:The authors have considered leader humility as a human aspect of a leader and examined the boundary conditions of knowledge sharing through leader humility. Results demonstrate that psychological safety promotes knowledge sharing as a mediator and is strengthened by leader competence and subordinates' OSE as relatively under-explored moderators.

Link: https://doi.org/10.1108/JKM-03-2025-0386

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Noni (Morinda citrifolia Linn.) Fruit Extract Protects Against Non-precipitated Alcohol Withdrawal Symptoms in Mice

AUTHOR :

A.; Chimakurthy, J.; Pandy, V.; R.

JOURNAL NAME:

Journal of Pharmacology and Pharmacotherapeutics

DETAILS:

Published on 20 December 2025

ABSTRACT:

Background Noni (Morinda citrifolia Linn.) fruit has a long history of traditional use for treating several health conditions, including symptoms of major depression and anxiety. Purpose This study focused on evaluating the therapeutic potential of the methanolic extract of Morinda citrifolia Linn. fruit (MMC) and its ethyl acetate fraction (EAMMC) against alcohol withdrawal-induced symptoms like convulsion, anxiety, and anhedonia in mice using handling-induced convulsion (HIC), elevated plus maze (EPM), open field test (OFT), marble burying test (MBT), and sucrose preference test (SPT), respectively. Materials and Methods Swiss albino mice (male) weighing 30–35 g were used. Animals were grouped as saline withdrawn (SW) control alcohol withdrawn (AW), control, positive control (AW + diazepam 1 mg/kg i.p.), and MMC- and EAMMC-treated groups (AW + MMC 500 and 1,000; AW + 50 and 100 mg/kg p.o.), respectively. All test groups intraperitoneally received 1.25 mL/100 g b.w. of (20% v/v ethanol + 0.1% v/v 4-methylpyrazole) except the SW group, which received saline (1.25 mL/100 g, i.p.) instead of ethanol twice a day for 3 days. HIC, OFT, EPM, MBT, and SPT were performed subsequently. Results These results reveal that MMC (500 and 1,000 mg/kg p.o.) and EAMMC (50 and 100 mg/kg p.o.) attenuated the handling-induced seizures (HIS), anxiety, and anhedonia in AW mice by facilitating the benzodiazepine-GABAAergic and serotonergic (5-HT1A) mediated mechanisms in the brain. Conclusion MMC and EAMMC protect against alcohol withdrawal symptoms such as handling seizures, anxiety, and anhedonia in mice.

Link: https://doi.org/10.1177/0976500X251398613

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Deep Learning Strategies for Optimizing Customer Segmentation

AUTHOR :

A.; I.; Guo, B.; R.; T.; Jung, Y.

JOURNAL NAME:

Americas Conference on Information Systems, AMCIS 2025

DETAILS:

Published on 2025

ABSTRACT:

Customer segmentation is essential for businesses to develop targeted strategies based on purchasing behavior. Traditional clustering methods, such as K-Means, have been widely applied in this area; however, they struggle with high-dimensional data and complex, non-linear customer relationships. To address these limitations, this paper explores the integration of deep learning techniques-specifically, autoencoders for dimensionality reduction-with conventional clustering algorithms. Autoencoders compress high-dimensional data into a more manageable format while preserving critical patterns, enabling more accurate and meaningful segmentation. This hybrid approach is applied to a large retail dataset, demonstrating its effectiveness in uncovering distinct customer groups with greater precision. The resulting segments offer deeper insights into customer preferences and behaviors, allowing businesses to refine their marketing campaigns, personalize customer experiences, and optimize product offerings. By leveraging deep learning alongside traditional clustering, companies can enhance decision-making and drive customer-centric growth in an increasingly data-driven market.

Link:
https://www.scopus.com/pages/publications/105025137552

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Nestlé's new recipe for success: a CEO's strategy amid inflation

AUTHOR :

Sainger, G.; Khan, W.

JOURNAL NAME:

CASE Journal

DETAILS:

Published on 4 December 2025

ABSTRACT:

Research methodology This case study was based on secondary sources of data to analyze Nestlé's strategic restructuring and turnaround strategies. The secondary sources included industry reports, company annual reports, press announcements, financial filings and business news stories from reputable sources such as Reuters, Forbes and Barron's. These sources offered in-depth observations of Nestlé's operational issues and strategic interventions in the times of weakening performance and market volatility. The case synthesized these observations with applicable strategic management theories – such as corporate restructuring, turnaround strategy, the BCG Matrix and PESTEL analysis – to deliver a rigorous but tractable model for classroom discussion. This secondary data-driven approach ensured factual dependability along with providing a sound basis for academic teaching strategy, organizational transformation and competitive positioning. Case overview/synopsis Nestlé the world's largest food company is experiencing decreasing sales, low operating efficiencies and increased competition. The stock value declined by 29% between January 2022 and August 2024. In August 2024, newly appointed CEO Laurent Freixe presented a turnaround strategy by restructuring operations, focusing on 31 "billionaire brands" and price competitiveness. Challenges remain in terms of economic uncertainties, inflation and geopolitical risks. This case assesses whether Freixe's strategy can regain the market leadership for Nestlé amid operational and market challenges. Complexity academic level This case study can be used for graduate-level and executive education programs, especially in strategic management, leadership and organizational change perspective.

Link: https://doi.org/10.1108/TCJ-01-2025-0014

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Developing a behavioral framework for sustainable entrepreneurship using interpretive structural modeling

AUTHOR :

Ingale, K.; Kulkarni, A.; Waghmare, H.

JOURNAL NAME:

Journal of the International Council for Small Business

DETAILS:

Published 16 December 2025

ABSTRACT:

The significance of sustainable practices in today's world is underscored by escalating concerns over environmental degradation and the socioeconomic challenges posed by climate change. A new class of entrepreneurs, known as sustainable entrepreneurs (SEs), has emerged amid these developments, which include green entrepreneurs, social entrepreneurs, and impact entrepreneurs, who aim to achieve both social and environmental benefits alongside financial return. This study examines the behavioral drivers that motivate SEs and presents a conceptual framework that illustrates the hierarchical relationships underlying decision-making processes in establishing sustainable businesses. The study employs a qualitative research design guided by an interpretive paradigm using interpretive structural modeling (ISM). It was observed that moral altruism, egocentric beliefs, and biospheric ideals formed the base layer of the final ISM model and influenced a maximum number of constructs, while the dependence power of intention was the highest. With the core values influencing higher-level constructs like norms and intention, the finalized ISM framework produced the hierarchical relationships among key variables influencing sustainable business intentions. These findings were corroborated through an expert interview. The study contributes toward theory development by integrating two grand theories; namely, the theory of planned behavior and value–belief–norm theory, to a specific context of sustainable entrepreneurship, which will pave the way for theory development at the meso level. However, the ISM model excludes institutional and socioeconomic factors. Such components could be added to the model in further studies to construct a comprehensive model of sustainable entrepreneurial behavior.

Link: https://doi.org/10.1080/26437015.2025.2596698

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Multidisciplinary Project Management for Organizational Success: Advanced Techniques for Engineering, Science and Management

AUTHOR :

Dubey, U.K.B.; Kothari, D.P.

JOURNAL NAME:

Book: Multidisciplinary Project Management for Organizational Success: Advanced Techniques for Engineering, Science and Management

DETAILS:

Published on October 2025

ABSTRACT:

Project management can play a key role in arriving at an effective decision in various functional areas of management. This book focuses on the integral role of project management in an organization. All aspects of a project, such as project identification, project planning and scheduling, project implementation, project evaluation, project financing and project audit, are covered in detail in this book.

Features:

• Covers in depth concepts and techniques of project management

• Emphasizes project management during the pre-stages of project development

• Provides keywords of essential vocabulary so students can recognize and retain important terminology

• Includes case studies taken from real-life situations showing the practicalities involved in the discipline

• Focuses on the practical aspects of project analysis and implementation as they relate to entrepreneurs

• Features glossaries of central term definitions and of frequently used symbols

It is designed to help undergraduates, postgraduates, and research professionals understand the basic concepts and applications of project management.

Link: https://doi.org/10.1201/9781315162997

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

The Proliferation of Artificial Intelligence in India: Legal and Regulatory Challenges in Light of Privacy and Deepfakes Concerns

AUTHOR :

Singh, D.; Pant, A.; Chaturvedi, V.

JOURNAL NAME:

In book: Moral and Legal Aspects of Artificial Intelligence

DETAILS:

Published on July 2025

ABSTRACT:

The rapid growth of Artificial Intelligence (AI) in India has led to improvements in sectors like healthcare, finance, and education. However, concerns about data privacy, security, and the potential for AI-generated content to be used for malicious purposes, such as deepfakes, have emerged. India's current legal and regulatory framework is fragmented and inadequate, lacking clarity, self-regulation, and effective enforcement mechanisms. This research aims to examine whether India's existing legal and regulatory framework is sufficient to address the challenges posed by AI proliferation, particularly in relation to privacy and deepfake concerns. A multi-pronged approach is recommended, including strengthening data protection, enacting targeted deepfake legislation, imposing transparency and accountability requirements, and adopting ethical safeguards.

Link: https://doi.org/10.4018/979-8-3373-3114-0.ch001

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Development of Mental Health Prediction App for the Depression Assistance Based on AI Chatbot

AUTHOR :

Singh, H.P.; Singh, N.; A.; P.K.; S.; K.; Sharma, N.

JOURNAL NAME:

Conference: 2025 International Conference on Engineering Innovations and Technologies (ICoEIT)

DETAILS:

Published on 31 October 2025

ABSTRACT:

Recently, the rise in mental health issues, especially depression, has significantly increased, requiring an additional support system. The objective of this work is to develop a mental health prediction app that provides support to individuals by integrating an AI-enabled chatbot. This app uses a client-server architecture with various modules and offers personalized treatment plans and self-care strategies based on evidence-based practices. It also includes features such as tracking and monitoring of mood, stress, and sleep patterns, peer support mechanism, and the option to connect with healthcare providers. The success of the app will depend on user engagement, adoption rates, privacy assurance, and data security. This work underscores the potential of mobile health (mHealth) Android apps to support mental health and well-being, offering a promising solution for enhancing accessibility and empowering individuals to manage their well-being. For this study, multi-tune transcript featuring dataset is utilized and simulated using python. The system performance is evaluated using standard metrics including accuracy, precision, recall, ROC etc. The proposed model achieved the accuracy of 93.5% which performs better than the traditional approach and demonstrates its effectiveness as a robust solution for mental health.

Link: https://doi.org/10.1109/ICoEIT63558.2025.11211786

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Computational Investigations Of Sound Transmission Loss In Hybrid Multi - Layered Fiber - Aluminium Composite Material

AUTHOR :

Vedpathak, S.G.; Chavande, Y.; Ghadge, R.R.

JOURNAL NAME:

Conference: International Mechanical Engineering Congress and Exposition - India Proceedings Series

DETAILS:

Published on 3 December 2025

ABSTRACT:

Noise pollution has caused problems for the health of people, workers, and the environment mainly in cities and factories for this reason, soundproof materials are essential in reducing indoor noise levels. For this research, the researcher evaluates the way glass fiber, basalt fiber, and perforated aluminum contribute to sound acoustics. The effectiveness of the composite design depends on fiber direction, the order of the sheet, and the pattern's cutting. When more layers are added, the performance and stability of the acoustics of the sound system get better. The perforated discs made from aluminum increase the release of sound through structural resistance. The advantage comes from glass and basalt fibers stopping heat and noise, and rubber helping to absorb vibrations both at the office and in cars. Furthermore, Lightweight acoustic models have the ability to perform highly. and great at keeping noise in. They greatly help the acoustics and lower the energy used. There is a need for scientists to carry on investigating and verifying the results in experiments to make the acoustics better, build sustainability, and improve the possibility of mass production. This paper describes the role of acoustic engineers using new methods and tools to make today's buildings and infrastructure quieter.

Link: https://doi.org/10.1115/IMECE-INDIA2025-159907

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Multi-Modal Biometric Authentication: Integrating Deep Learning to Enhance Identity Theft Protection

AUTHOR :

Selvi, V.; Bader Alazzam, M.B.; Sulaiman, S.F.; Majid, A.-H.M.; Kakad, S.; Rajasree, S.

JOURNAL NAME:

Conference: 2025 International Conference on Engineering Innovations and Technologies (ICoEIT)

DETAILS:

Published on 31 October 2025

ABSTRACT:

Identity theft is still a severe issue, and single-modal biometric systems (such as fingerprint or facial recognition) can be spoofed and become inaccurate under different conditions. As an antidote to these weaknesses, this work presents a CNN-RNN-based multi-modal biometric authentication system that uses face, fingerprint, and voice information for added security and accuracy. CNNs derive spatial features (e.g., face patterns), whereas RNNs analyze temporal patterns (e.g., voice signals), allowing the system to acquire sophisticated biometric features. In contrast to other conventional models like Random Forest and Logistic Regression, which exhibited subpar performance with recognition rates below 90% and low capacity for handling multi-modal data, our solution has a recognition rate of 99.3%, precision of 98.4%, and F1-measure of 97.3%. This has been shown to be a tremendous leap forward from current systems in that it captures both temporal and spatial features. The suggested deep learning-based framework provides an efficient, reliable, and scalable solution to present biometric authentication problems.

Link: https://doi.org/10.1109/ICoEIT63558.2025.11211651

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Role and Influence of Artificial Intelligence (AI)-Powered Technologies in Terms of Enhancing Customer Satisfaction in the Hospitality Industry

AUTHOR :

Chaudhary, P.; Sharma, S.; Chanda, R.

JOURNAL NAME:

Journal of Information and Knowledge Management

DETAILS:

Published on December 2025

ABSTRACT:

With the advent of Industry 4.0, the use of Artificial Intelligence (AI) and digital technologies has become indispensable for the hospitality industry. This study focused on how AI in the hospitality industry has enhanced customer experience and satisfaction, along with other factors. The study has adapted models such as SERVQUAL and ECSI from the literature, proving the relation between AI technologies and customer satisfaction. Responses from 276 participants were collected and analysed using confirmatory factor analysis and structural equation modelling. This study found that AI had a significant relation to enhancing customer satisfaction (and thereby overall experience). The perceived value affects customer satisfaction, which is influenced by the expectancy and the threat of risk. This study essentially delves deep into consideration response time and the threat of risk, which has emerged as one of the sticking points after the upsurge in the use of AI in the hospitality sector.

Link: https://doi.org/10.1142/S0219649225501266

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Characterization of aggregate concrete with eco-friendly silicon additives: a green material for sustainable infrastructure

AUTHOR :

Shastri, D.S.; Kumar, S.; Singh, A.K.; Rawat, M.S.; Manju, J.; L, L.; Geetha, S.; Babu, L.G.

JOURNAL NAME:

Composite Interfaces

DETAILS:

Published on 16 December 2025

ABSTRACT:

This research investigates the influence of rice husk ash (RHA) and biosilica (BS) addition on the workability and mechanical properties of aggregate concrete. Concrete mixtures incorporating 2, 3 and 5 vol. % of RHA and BS were compared against a control mix. Further, workability tests revealed that the inclusion of RHA reduced slump significantly with the greatest decline observed at 5 vol. % (23.3% reduction), due to its porous and absorptive nature. Conversely, biosilica exhibited better slump retention, with only a minor 4.2% reduction at 2 vol. % addition. Additionally, compressive strength (CS) results demonstrated that the optimal improvements were achieved at 2–3 vol. % replacements. At 28 days, RHA mixes attained strengths up to 47.3 MPa, while BS mixes achieved higher values, peaking at 50.1 MPa. Excessive substitution (5 Vol. %) led to strength reductions, more pronounced in RHA-based mixes. Moreover, split tensile strength (STS) followed a similar trend. At 56 days, the highest STS was obtained for B3 (4.6 MPa), surpassing R3 (4.2 MPa) and the control (3.2 MPa). The results confirm that while both RHA and BS contribute to strength enhancement due to its finer particle size, higher amorphous silica content and greater reactivity.

Link: https://doi.org/10.1080/09276440.2025.2597690

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Hybrid Artificial Intelligence-Based Approaches for Rainfall Forecasting

AUTHOR :

Yadav, A.; Singh, J.; Jayshree; Joshi, D.; Pradhan, A.K.

JOURNAL NAME:

Conference: 2025 International Conference on Artificial intelligence and Emerging Technologies (ICAIET)

DETAILS:

Published on 30 October 2025

ABSTRACT:

With the rise in need of Electronic Health Records (EHR), keeping confidentiality, integrity, and privacy intact of Patient Health Information (PHI) becomes of utmost importance, particularly while sharing via email which is inherently non-secure. The current work proposes a hybrid encryption approach on the basis of Advanced Encryption Standard (AES-256) with symmetric key cryptography and Elliptic Curve Cryptography (ECC) as asymmetric encryption. AES-256 is selected because it performs well to encrypt large datasets, and ECC provides secure key exchange and digital signature support with less computation overhead compared to RSA. The system addresses issues of key encryption, such as secure transmission of information, identification authentication, and digital signatures for message integrity.Besides AES and ECC, the system incorporates a Zero-Knowledge Proof (ZKP) protocol, enabling privacy-preserving identity authentication without exposing sensitive data. This is important in ensuring PHI protection because ZKP permits identity and message authenticity verification with private data left secure. The hybrid system is balanced between speed of encryption and security, such that it remains scalable and usable for real-time healthcare use cases. By integrating AES for quick data encryption, ECC for secure key management, and ZKP for added privacy, this solution provides an end-to-end solution for securely sending PHI through email. It successfully solves encryption issues while maintaining confidentiality, integrity, and privacy in healthcare communication without compromising efficiency.

Link: https://doi.org/10.1109/ICAIET65052.2025.11210929

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Adaptive PID Tuning for Quadcopters in Dynamic Environments Using Advantage Actor-Critic Reinforcement Learning

AUTHOR :

Omkar, S.N.; Maski, K.; S.

JOURNAL NAME:

Conference: 2025 International Conference on Emerging Technology in Autonomous Aerial Vehicles (ETAAV)

DETAILS:

Published on 29 October 2025

ABSTRACT:

Achieving precise and adaptive control of quadcopter systems is essential for aerial robotics, autonomous navigation, and dynamic environment interaction. This study investigates the implementation of the Advantage Actor-Critic (A2C) algorithm, a prominent deep reinforcement learning (DRL) technique, for the online optimization of Proportional-Integral-Derivative (PID) controllers. The proposed framework autonomously adjusts PID gains through real-time interaction with a simulated environment, facilitating adaptive control under highly dynamic and uncertain conditions. Rigorous validation was conducted using simulated flight paths defined by discrete waypoints, designed to emulate complex operational scenarios. These scenarios incorporated varying payloads, external disturbances such as environmental noise, and significant mass uncertainties, illustrating the robustness of the approach. By utilizing a continuous state-action space, the A2C algorithm ensures effective adaptation and stability within the multi-input multi-output (MIMO) dynamics inherent to quadcopter control. The experimental results underscore the potential of DRL-enhanced control mechanisms for real-time, noise-resilient operation, offering a pathway toward increased autonomy, reliability, and efficiency in aerial systems.

Link: https://doi.org/10.1109/ETAAV66793.2025.11213110

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Nonlinear thermo-mechanical bending analysis of laminated composite plates by using trigonometric shear deformation theory

AUTHOR :

S.R.; Ghugal, Y.M.; Sayyad, A.S.

JOURNAL NAME:

Mechanics of Advanced Materials and Structures

DETAILS:

Published on 4 December 2025

ABSTRACT:

The paper presents the analytical solutions for thick orthotropic laminated plates using refined shear deformation theory. The effects transverse shear and transverse normal strains are included. The displacement field of the theory includes the trigonometric functions in thickness coordinate of plate to account for these effects. The displacement field enforces to give the realistic variation of shear stress across the thickness of plate and thus obviates the need of shear correction factor. The principle of virtual work is used to obtain the governing equations and boundary conditions. Simply supported thick square laminated plates are considered for numerical study. The results obtained by present theory are compared with other refined theories available in the literature. It is observed that the results obtained from present theory are in good agreement with other higher-order shear deformation theories.

Link: https://doi.org/10.1080/15376494.2025.2588806

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Optimization of internal geometry for maximum transmission loss in a reactive muffler

AUTHOR :

Kulkarni, S.; Kulkarni, M.V.

JOURNAL NAME:

Building Acoustics

DETAILS:

Published on 11 December 2025

ABSTRACT:

Noise pollution from internal combustion engines in vehicles and industrial equipment poses an environmental challenge. Exhaust systems use mufflers to reduce noise by attenuating sound waves. This study focuses on optimizing the extended inlet length and diameter in a reactive muffler with a single expansion chamber and extended sections. The aim is to enhance transmission loss in a targeted frequency range for better noise reduction. Using the Taguchi Method, the study identified optimal design parameters for sound attenuation. Acoustic performance was assessed through two methods: (1) numerical simulations using COMSOL Multiphysics and the Finite Element Method (FEM) to analyze sound wave behavior, and (2) experimental validation via the two-load method to measure transmission loss in a prototype. Results highlight key design insights for optimizing reactive mufflers, offering a robust framework for achieving superior noise control in applications requiring frequency-specific attenuation.

Link: https://doi.org/10.1177/1351010X251389683

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Optimization Methods and Algorithms

AUTHOR :

Kulkarni, A.J.

JOURNAL NAME:

Book: Optimization Methods and Algorithms

DETAILS:

Published on January 2025

ABSTRACT:

The book thoroughly explains fundamental optimization concepts and terminology, including variables, parameters, constraints, bounds, and convexity. It also demonstrates how to formulate optimization problems using illustrative examples. Covering both single-variable and multi-variable optimization methods, the book provides theoretical insights, practical examples, and exercises, along with a graphical approach to problem-solving.

In light of growing concerns about resource limitations and environmental impacts, this textbook addresses the need for efficient resource use amidst technological advancements and market competition. Students will appreciate the comprehensive coverage, supported by illustrations and exercises that deepen their understanding. Instructors will find it invaluable for classroom teaching, with accessible concepts and practical examples that highlight the nuances of optimization.

Link: https://doi.org/10.1007/978-981-96-9194-4

Since 1983

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Activity-centric risk analysis for delay prediction in Indian tunnel projects

AUTHOR :

Shelake, A.G.; Gogate, N.G.; Khatu, D.R.

JOURNAL NAME:

Civil Engineering and Environmental Systems

DETAILS:

Published on 11 December 2025

ABSTRACT:

Indian tunnel projects often face delays that inflate budgets, disrupt timelines, and threaten project success. This study introduces a novel activity-centric, scenario-based risk analysis framework to forecast and mitigate delays by combining risk parameters: probability, exposure, and consequence. Delay predictions are made by analysing risk scenarios probability-exposure (Scenario 1), exposure-consequence (Scenario 2), and probability-consequence (Scenario 3). The methodology starts by linking 16 most significant risk factors to tunnel activities. Activity-wise risk levels are computed using an AHP-driven Event Tree Analysis (ETA) to evaluate consequences and Fault Tree Analysis (FTA) to assess probability and exposure. These levels are integrated into the project schedule to forecast delays. Methodology enables systematic identification, prioritisation, and tracking of cascading risk impacts at the activity level. Drawing real-time data from Indian metro tunnel projects, the study reveals distinct risk scenarios are more effective for the planning and execution phases. Scenario 3 (exposure-consequence) closely aligns with delays in planning, while scenarios 1 and 2 perform better for execution. Findings reveal that major delay-sensitive activities include NATM excavations, TBM dragging, method-statement preparation, and soil investigation. The study also introduces an activity-based risk matrix with tailored mitigation strategies, strengthening schedule resilience and enabling project managers for timely, risk-informed project delivery.

Link: https://doi.org/10.1080/10286608.2025.2598569

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Determinants of India's groundnut export performance: an empirical analysis

AUTHOR :

Yadav, A.K.; Chattopadhyay, U.

JOURNAL NAME:

International Journal of Emerging Markets

DETAILS:

Published on 27 November 2025

ABSTRACT:

Purpose The growing importance of India's groundnut export in the world market stimulates research into this product. This article aims to econometrically investigate the impact of critical determinants on India's groundnut export performance over the long run and the short run. Design/methodology/approach This study employed an autoregressive distributed lag (ARDL) model among the considered variables using annual time series data spanning over 3 decades from 1990 to 2020. Findings The results confirm that yield, production cost and trade openness have positive significant impact, while the export price has negative significant impact on groundnut export performance in the long run. In the short-run, market size, last year's yield, last year's trade openness and the current year's export price have significant negative impacts on India's groundnut exports. However, the findings of the study revealed that the current year's trade openness and last year's export price have a significant positive impact on India's groundnut export performance in the short run. Research limitations/implications The findings of this study offer some important policy implications for farmers, exporters and the government by indicating the key factors that can help improve the export performance of groundnut in the global market. Originality/value To the best of the author's knowledge, this research is the first of its kind that deploys the ARDL modelling approach to identify the factors that influence groundnut exports. Empirically, the study contributes to the existing literature by providing new insights on India's groundnut export performance. This research will also serve as a benchmark for assessing the product-level export performance of other agri-products.

.Link: https://doi.org/10.1108/IJOEM-08-2024-1375

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Global Financial Crises: Lessons From Policy Failures and Successes

AUTHOR :

Patil, A.A.; Kumar, M.; Sengupta, R.; Yadav, R.; Dixit, P.

JOURNAL NAME:

In book: Policy Implications on International Financial Economics and Banking

DETAILS:

Published on October 2025

ABSTRACT:

Financial crisis have always fascinated and concerned acadmicians, finance enthusiasts and policymakers alike. and This article explores the major global financial crises, including the Great Depression, the Latin American Debt Crisis, the Asian Financial Crisis, the Global Financial Crisis and the recent COVID-19 economic shock. The causes, policy responses, and outcomes of these crises, have been discussed to provide policy lessons in the context of financial crisis. T Key policy failures such as rigid monetary policies, austerity, and regulatory gaps exacerbated these economic crisis and the corresponding social damage, while .On the other hand, timely measures such as countercyclical stimulus, financial reforms, and global cooperation have facilitated recoveries during these times. This article synthesizes these insights into actionable principles for future crisis management, emphasizing swift liquidity provision, robust regulation, inclusive policies, and multilateral coordination.

Link: https://doi.org/10.4018/979-8-3373-3725-8.ch003

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Precision Medicine: Market and Its Potential in the Healthcare Industry

AUTHOR :

Kulkarni, M.; Sharma, M.

JOURNAL NAME:

Book: Digital Transformation and Artificial Intelligence for Operational Excellence in Healthcare

DETAILS:

Published on 2025

ABSTRACT:

Customized treatment plans based on environmental, lifestyle, and genetic principles are presently revolutionizing health care. This chapter traces the evolution of precision medicine (PM), emphasizing some of the basic principles and ever-expanding impact that the PM field exerts on modern-day medical practice. PM aims at maximizing therapeutic response while minimizing side effects and improving diagnostic accuracy, which can potentially yield benefits for the individual patient and the healthcare systems in general. With a global push for commercialization, technological advancement, and innovation funding from major industries, PM has been growing at an accelerated rate. The insights on market segmentation and regional analyses show the differences in growth patterns and investment opportunities among diverse healthcare systems. Advanced technologies, such as Artificial Intelligence (AI), omics sciences, implantable biosensors, big data analytics, and Internet of Things (IoT), contribute substantially to PM. Enabled by these advancements, precision health care now has new possibilities with data-driven decision-making, predictive analytics, and customized therapeutics. The journey into the future of PM will witness the formulation of novel treatments based on advances in personalized drug development, aided diagnostics, and genetic discoveries; the flipside of the PM story will be the data security, ethical issues, and regulatory compliance that must be looked into to ensure its long-standing acceptance. This chapter studies the commercial dynamics, technological advancements, and possible future directions in PM, and these elements are vital for every stakeholder willing to expedite improvements in patient outcomes and individualized health care.

Link: https://www.emerald.com/books/edited-volume/18206/chapter-abstract/101524848

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

The farmer's path to market: insights on extension services support, types of crops and selling practices from situation assessment survey of Indian farmers

AUTHOR :

Sehgal, V.; Biradar, J.

JOURNAL NAME:

Indian Growth and Development Review

DETAILS:

Published on 19 November 2025

ABSTRACT:

Purpose This study aims to understand the relationship between extension services, crop types and farmers' market participation in India. It further explores the factors influencing farmers' choice of market and the role of extension service in shaping their decision of market participation. Design/methodology/approach The multivariate logistic regression and probit model are used to understand factors influencing the farmers' choice of market participation. Findings This study finds that the major crop selling takes place through the non-regulated markets. This highlights the crucial role of non-regulated spaces in the Indian agricultural marketing system, in spite of their shortcomings. The same pattern can be observed across all crop categories, indicating that institutional market mechanisms are either inaccessible or inefficient for the majority of the farmers. The study further reveals that the private extension services have emerged as a key and widely relied upon source of market-related information. The results show that farm households mainly rely on private extension services for agriculture-related advice. The logistic regression shows that demographic, farm and institutional variables also significantly influence farmers' choice of market channels. The findings offer managerial implications by advocating for a dual market approach that improves transparency in non-regulated markets while addressing their inherent inefficiencies. This study also advocates for the strengthening of extension services to enhance farmer participation. Originality/value This study uniquely links extension services and crop choices to farmers' market participation, offering fresh evidence from India's unit-level data of the latest round (77th round, 2018–19) of the National Sample Survey on agricultural households in India.

Link: https://doi.org/10.1108/IGDR-06-2025-0102

TITLE:

Nanofillers in Energy Devices for Storage

AUTHOR :

Naik, J.; Kahane, S.; Nandimath, M.

JOURNAL NAME:

In book: Handbook of Nanofillers

DETAILS:

Published on 6 August 2025

ABSTRACT:

This chapter discusses the use of nanofillers in the energy field. Nanofillers, including materials such as carbon nanotubes, graphene, and clay nanoparticles, have unique properties that make them useful in a variety of energy-related applications. For example, they can be used as catalysts, electrodes, and thermal conductors in fuel cells, batteries, and solar cells. Additionally, they can be used as reinforcement materials in composites to improve the strength and thermal stability of structural components. The chapter reviews the current state of research on the use of nanofillers in the energy field, highlighting key findings and identifying areas for future research. It also covers the potential benefits and challenges associated with the use of nanofillers in energy-related applications and provides an overview of the current commercial developments and ongoing research in this field.

Link: https://doi.org/10.1007/978-981-96-2407-2_125

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Nanofillers in Drug Delivery Industry: Recent and Incoming Trends

AUTHOR :

Choudhari, V.; Satish, S.; Choudhari, G.; Topare, N.S.

JOURNAL NAME:

In book: Handbook of Nanofillers

DETAILS:

Published on 6 August 2025

ABSTRACT:

Nanofillers (NFs) are used in various pharmaceutical formulations to achieve desired drug delivery system (DDS) properties for successful drug delivery goals; to win over their intrinsic limitation of DDS. By nature, NFs have diverse characteristics, are conveniently prepared, and are cost-effective. Nanofillers includes organic, inorganic, carbon, nanocomposite materials and materials from other categories. Most of the nanofiller are lighter, durable, versatile, and are reliable for various applications. New types of materials such as metal oxide framework, 3D bioprinted, MXenes, and hydroxy apatite, are emerging, with excellent drug delivery applications. Nanofiller applications in various DDS such as hydrogels, tablets, wound dressings, films and membranes, microneedles, soft nongelation capsules, and microparticles are reviewed. Major applications are reported for hydrogels and transdermal delivery systems. Incorporation of NFs enable smart drug delivery to modulate release such as pulsatile release, improved release, and sustaining the release achieved. Smart drug deliveries such as dual responsive, triggerable, on demand, shape memory, enhanced drug encapsulation are enabled by NFs. Nanofillers preprepared using different methods, which have an effect on its properties, which in turn have an effect on DDS performance. Therefore, control of its physicochemical properties is essential. Various techniques used for characterization of the NFs such as spectroscopic, thermal, and advanced microscopy. Various aspects of NF use in hydrogel and films/membrane DDS are mentioned in brief. Future scope for NFs in DDSs and possible limitations are discussed. Regulatory approvals are essential and the relevant global scenario is discussed in brief.

Link: https://doi.org/10.1007/978-981-96-2407-2_83

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Electrical Properties of Polymer Nanocomposites Containing Rod-Like Nanofillers

AUTHOR :

Kahane, S.; Naik, J.

JOURNAL NAME:

In book: Handbook of Nanofillers

DETAILS:

Published on 6 August 2025

ABSTRACT:

The electrical characteristics of polymer nanocomposites incorporating rod-like nanofillers are briefly discussed in this chapter. A lot of interest has been paid to polymer nanocomposites containing rod-like nanofillers because of their distinct electrical characteristics and possible use in many different domains. Flexible electronics, sensors, energy storage, and electromagnetic shielding are just a few of the new possibilities made possible by the electrical properties of polymer nanocomposites with rod-like nanofillers. The distribution and alignment of rod-like nanofillers inside the polymer matrix have a significant impact on the electrical behavior of polymer nanocomposites. These nanofillers may be effectively dispersed and aligned to form percolation networks, which will facilitate charge transport channels and increase electrical conductivity. One of the major benefits of rod-like nanofillers is their high aspect ratio, which makes it easier to build conductive networks inside the polymer matrix. This network formation makes it simpler to generate efficient charge transport channels than neat polymers, which enhances electrical conductivity. The electrical performance of the nanocomposite is enhanced by the multiple pathways for electron or ion transport provided by the firmly connected nanofillers. The concentration, aspect ratio, and dispersion of the nanofiller within the polymer matrix can all be changed to alter the electrical conductivity of polymer nanocomposites. With the proper choice and modification of these parameters, a wide range of electrical conductivity values, from insulating to highly conductive regimes, can be obtained. Additionally, the electrical characteristics are greatly influenced by the interfacial contacts between the rod-like nanofillers and the polymer matrix. The interfacial adhesion and enhanced charge transfer between the nanofillers and the polymer matrix can be improved by surface modifications of the nanofillers or chemical compatibilizers. Furthermore, adding rod-shaped nanofillers to polymer matrices improves dielectric characteristics, mechanical strength, thermal stability, etc.

Link: https://link.springer.com/rwe/10.1007/978-981-96-2407-2_34

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Nanoparticles and Nanofillers in Drug Delivery Industry

AUTHOR :

Wagh, S.S.; Chougale, A.S.; Shelake, H.D.; Topare, N.S.; Gunnasegaran, P.; Syed, A.

JOURNAL NAME:

In book: Handbook of Nanofillers

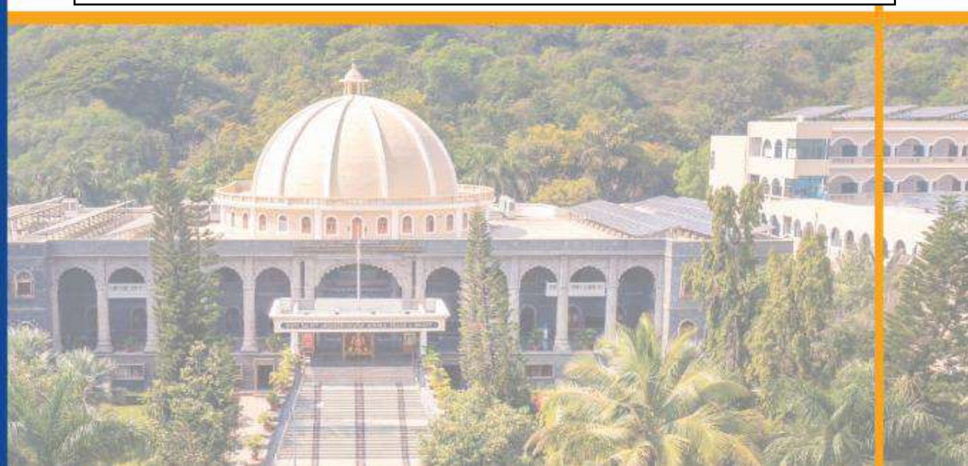
DETAILS:

Published on 6 August 2025

ABSTRACT:

Recently, nanofillers have received great attention due to their versatile characteristics-especially those based on carbon nanotubes and clays. Nanographene platelets and montmorillonite clay are communal examples of one-dimensional (1D) nanoparticles. The well-known definition of nanofillers is that they must have at least 1D in the range between 1 and 100 nm and the others should be bigger than 100 nm. They are generally in the form of sheets of 1 nm to a few nanometers thick to hundreds and thousands of nanometers long. Nanofillers as a group are barely considered unique anymore, meanwhile, various products hold them but they still attract innovative research, particularly in the area of graphene and further carbon-related fillers. Nanofillers can be categorized into three types: one nanoscale dimension (nanoplatelet), two nanoscale dimension (nanofiber), and three nanoscale dimension (nanoparticulate). The advantage of such nanofillers is that they produce very good mechanical properties at low loadings, enhanced fire-resistant properties, superior barrier properties, scratch resistance, and improved heat distortion performance when compared to pristine material. The main objective of this chapter is to review the research on various types of nanofillers and their effect on the structural, morphological, and optical properties of pristine base materials and also its performance in various applications.

Link: https://doi.org/10.1007/978-981-96-2407-2_84



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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

The Impact Of Artificial Intelligence On Qbd Implementation In Analytical Laboratories: A Comprehensive Review

AUTHOR :

Nair, S.; Pishorkar, G.; Satish, S.

JOURNAL NAME:

Pharma Times (Vol.-57, Issue-9)

DETAILS:

Published on 2025

ABSTRACT:

In contrast to rule-based systems and individuals, deep learning and artificial intelligence can filter through huge amounts of data to discover intricate patterns. Testing in the laboratory serves several purposes in the field of laboratory medicine, which is not surprising considering the significance it plays in the process of clinical decision-making. Artificial intelligence and machine learning have already increased utilisation, automated laboratory processes, and offered individualised reference ranges and test interpretations. This is even though many applications are still in their very early stages of development. As a consequence of the findings of the current inquiry, practitioners in the laboratory will progressively use machine learning and artificial intelligence. Patients and medical professionals alike are now able to make use of artificial intelligence (AI) in laboratory settings. The scientific community has high expectations that these discoveries will, in the future, make diagnostic work performed in laboratories more sensitive and effective due to their increased sensitivity.

Link:
https://www.scopus.com/pages/publications/105024359422

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Nanomedicine A Promising Way to Manage Alzheimer's Disease

AUTHOR :

Pawar, S.; Nayak, P.G.; Sharma, S.; Singh, S.; Puranik, N.; Sharma, A.

JOURNAL NAME:

In book: Nanomedicine in the Treatment and Management of Alzheimer's Disease

DETAILS:

Published on January 2025

ABSTRACT:

Alzheimer's disease (AD) is a neurodegenerative disorder marked by cognitive decline and memory loss. Traditional therapeutic approaches have shown limited success, necessitating innovative strategies. Nanomedicine, the application of nanotechnology in medicine, presents a promising avenue for managing AD. This emerging field leverages nanoparticles to improve drug delivery, enhance diagnostic capabilities, and facilitate targeted therapy. Nanoparticles, including liposomes, dendrimers, polymeric nanoparticles, and metallic nanoparticles, offer unique advantages due to their small size, large surface area, and ability to cross the blood–brain barrier (BBB). Recent advancements in nanomedicine have focused on developing nanoparticles for amyloid-beta (Aβ) plaque targeting, tau protein aggregation inhibition, and neuroinflammation reduction. Functionalized nanoparticles can selectively bind to Aβ plaques, promoting their clearance and reducing amyloid burden. Similarly, tau-targeted nanoparticles can interfere with tau aggregation, potentially mitigating tauopathies. Nanomedicine also holds promise for early diagnosis and monitoring of AD progression. Nanoparticles conjugated with imaging agents enable high-resolution imaging of amyloid plaques and tau tangles, facilitating early detection and longitudinal studies of disease progression. Nanomedicine offers a multifaceted approach to managing AD by improving drug delivery, enhancing therapeutic targeting, and enabling early diagnosis. Continued research and development in this field hold significant potential for advancing AD treatment and improving patient outcomes.

Link: https://doi.org/10.1201/9781003543381-5

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Metaheuristic Methods for Structural Optimization: A Machine-Generated Literature Overview

AUTHOR :

Kale, I.R.

JOURNAL NAME:

Book: Metaheuristic Methods for Structural Optimization

DETAILS:

Published on June 2025

ABSTRACT:

This book presents the result of an innovative challenge, to create a systematic literature overview driven by machine-generated content. This machine-generated volume, with chapter introductions by the human expert, of summaries of the existing studies furthers our understanding of the heuristic and metaheuristic optimization methods for structural engineering domain problems. It brings out nature inspired metaheuristic optimization methods such as Genetic Algorithm, Particle Swarm Optimization, Ant Colony Optimization, Cohort Intelligence, Firefly Algorithm, Differential Evolution, Bee Colony, Grey Wolf Optimizer, League Championship Algorithm, Harmony Search Algorithm, Water Cycle Algorithm, Neural Network Algorithm, and its different variations that have been widely used to solve structural engineering problems.This book will be helpful for academicians and researchers in many kinds such as surveys of nature inspired optimization algorithms for structural optimization, its applicability for solving single objective and multi-objective structural engineering problems, constraint handling, solution quality, challenges, opportunities, key features, and limitations.Questions and related keywords were prepared for the machine to query, discover, collate and structure by Artificial Intelligence (AI) clustering. The AI-based approach seemed especially suitable to provide an innovative perspective as the topics are indeed both complex, interdisciplinary and multidisciplinary. Springer Nature has published much on these topics in its journals over the years, so the challenge was for the machine to identify the most relevant content and present it in a structured way that the reader would find useful. The automatically generated literature summaries in this book are intended as a springboard to further discoverability. They are particularly useful to readers with limited time, looking to learn more about the subject quickly and especially if they are new to the topics.

Link: https://link.springer.com/book/10.1007/978-981-95-1711-4

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

An Automatic Event Detection from Social Media Using an Improved Meta-Heuristic Model-Based Multi-Scale Deep Learning Network

AUTHOR :

Patankar, R.; Pravin, A.

JOURNAL NAME:

Journal of Information and Knowledge Management

DETAILS:

Published on October 2025

ABSTRACT:

Social media has become the primary and popular platform to share incidents that happen to every individual. Hence, the data generated by social media represents richer information by considering the ongoing events. Further, the timeliness associated with the data can provide valuable insights. Social media is one of the active communication models for connecting with each other, so event detection on social media is a widespread concern. Moreover, the deep learning models are introduced for detecting events based on temporal and spatial representations. Yet, focusing on the dynamic and large volume of data available in social media is impractical for filtering the events manually, and it is a time-consuming process. Further, the pre-trained models are less effective because of considering the mis-spelt words and enabling serious information loss. To alleviate these drawbacks in the conventional models, the research study employs a novel deep learning model for detecting the events in the social media effectively. This research proposed a multi-scale deep learning for event detection by analysing the social media data. The news agents or the decision maker can analyse the insight into the situation with the proposed event detection model. In the beginning, significant data are collected from publicly available datasets. Further, the gathered data is given to the text pre-processing stage for removing the redundancies to generate clean and quality data. Further, the pre-processed text is fed into the Optimal Weighted Multi-scale Transformer-based Deep Network (OWMT-DNet), in which the deep network is taken as Residual Bi-directional Long Short-Term Memory (Res-Bi-LSTM) for social event detection. In the proposed network, the weight optimisation is done by using the proposed algorithm named Opposition Piranha Foraging Algorithm (OPFA). Finally, the experimental validation is done for the recommended event detection model and compared with baseline approaches.

Link: https://doi.org/10.1142/S0219649225501151

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Novel Isocratic HPLC Method for Simultaneous Estimation of Sacubitril and Valsartan in Bulk and Formulation

AUTHOR :

Karnik, H.R.; Kuchekar, A.B.; Marne, A.B.; Gawade, A.R.

JOURNAL NAME:

Brazilian Archives of Biology and Technology (Vol.-68)

DETAILS:

Published on January 2025

ABSTRACT:

In compliance with ICH requirements, a new selective, precise, and accurate reverse-phase High-Performance Liquid Chromatographic method for valsartan and sacubitril quantification was developed and validated. Eluents were detected at 284 nm. The HPLC method was designed utilizing a C18, (150mm × 4.6mm, 3.5µm) column with 1% Acetic acid in water: Methanol (45:55 v/v) as a mobile phase at a flow rate of 0.8 mL/min. For valsartan and sacubitril, the calibration curves were linear over the concentration range of 10 to 100 ng/mL (R2 = 1) and 20 to 140 ng/mL (R2 = 1), respectively. Valsartan and sacubitril were shown to have average retention times of 5.79 and 7.019 minutes, respectively. Valsartan and sacubitril had average percentage recoveries of 99.83± 0.14 and 100.06± 0.19 percent, respectively. Valsartan and sacubitril had respective LOD values of 0.0958 and 0.1059 ng/mL. The suggested method's intraand inter-day precision values (% RSD) were under 2%. For the simultaneous measurement of valsartan and sacubitril, a straightforward, exact, accurate, linear, and quick RP-HPLC technique was created and verified by ICH recommendations. The findings imply that the devised approach can be used for pharmaceutical formulation and routine bulk measurement of valsartan and sacubitril. The method's applicability was confirmed through successful analysis of pharmaceutical formulations, indicating its potential utility for routine quality control and stability testing of Sacubitril and Valsartan in combined dosage forms. The choice of gradient elution, despite its complexity, was justified by the need for improved separation and resolution, which are critical for the accurate determination of these therapeutic agents.

Link: https://doi.org/10.1590/1678-4324-2025240353

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Microbial Synthesis of Melanin by Bacillus proteolyticus MITWPUME1: Isolation, Characterization, and Potential Application

AUTHOR :

Gaonkar, V.V.; Shinde, S.S.; Mukadam, H.M.; Sayyed, F.A.; Sasane, K.S.; Gaikwad, S.V.

JOURNAL NAME:

Research Journal of Pharmacy and Technology (Vol.-18, Issue-10)

DETAILS:

Published on October 2025

ABSTRACT:

Melanin pigment has diverse chemical and biological functions, that makes this molecule vital for potential applications in various fields. Microbial synthesis of melanin offers numerous advantages over chemical methods, including sustainability and cost-effectiveness. In the presented study, Bacillus proteolyticus MITWPUME1, an isolate of the Mimosa pudica plant's rhizospheric zone was found to be a potential producer of melanin, yielding 355 g/L with L-tyrosine supplemented media. Extracted melanin exhibited its peak absorption at wavelength 220 nm. The FTIR peaks' spectrum confirmed the presence of biomolecule-melanin. The biomolecule showed solubility in sodium hydroxide, ammonium hydroxide, and chloroform, while found to be insoluble in water, ethanol, and methanol. It shows precipitation with hydrochloric acid and decolorization with hydrogen peroxide. On the other hand, it showed stability across a broad range of pH and temperature, indicating its compatibility with a diverse range of industrial applications. The Sun Protective Factor (SPF) was found to be 6.15 and antioxidant activity 18.95%. These two important properties highlight the applicability of the biomolecule-melanin in the field of cosmetics. Thus, Bacillus proteolyticus MITWPUME1 emerges as a promising candidate for melanin producer with desired physicochemical properties, holding promise for other diverse agricultural, and biomedical applications.

Link: https://doi.org/10.52711/0974-360X.2025.00700

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Human–computer interaction for cognitive, emotional and learning well-being

AUTHOR :

Upreti, K.; Kumari, P.; Akilbasha, U.; Radhakrishnan, G.; Udhaya, S.K.; Malik, K.

JOURNAL NAME:

Book: Intelligent Systems for Neurocognition and Human-Robot-Computer Interaction

DETAILS:

Published on 2025

ABSTRACT:

Human–Computer Interaction (HCI) has revolutionized the way humans engage with technology, shaping cognitive, emotional, and learning experiences. This chapter explores HCI's impact on well-being, focusing on cognitive load reduction, emotional stability, and adaptive learning. HCI technologies such as AI-driven decision support, emotion-aware systems, and personalized education platforms enhance user engagement by fostering efficiency and well-being. Cognitive well-being benefits from AI-powered cognitive tools that improve memory, decision-making, and mental agility. Emotional well-being is facilitated by affect-sensitive systems, digital therapeutics, and social HCI that reduce stress and increase emotional engagement. Adaptive learning systems, gamification, and assistive technologies also ensure inclusive education by making learning more personalized and accessible to special needs students. Future HCI trends involve neuroadaptive interfaces, wearable-integrated health technology, and AI-based mental health solutions that all improve personalization and user experience. Yet ethical issues, such as data privacy, digital addiction, and algorithmic biases, need to be addressed in order to maintain responsible technology utilization. Achieving balance between digital and physical interactions is important in the preservation of general well-being. The future of HCI, where it becomes empathetic, adaptive, and ethical interfaces, is a future that sees technology enhance not just efficiency but cognitive, emotional, and learning well-being. By making ethical AI design and user-centric experiences a priority, HCI will continue to unleash human potential and develop durable, well-being-oriented technological solutions.

Link: https://doi.org/10.1016/B978-0-443-41660-6.00008-9

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Theranostics in bone cancer

AUTHOR :

Soni, U.; Thorat, V.G.; Pujari, R.

JOURNAL NAME:

In book: Theranostics in Cancer Management

DETAILS:

Published on October 2025

ABSTRACT:

Bone cancer, although rare, poses significant challenges in diagnosis and treatment due to its complex nature and the heterogeneity of tumors. Theranostics allows for personalized treatment plans tailored to the unique characteristics of each patient's cancer, utilizing molecular imaging techniques such as positron emission tomography (PET) to visualize cancerous tissues through specific biomarkers. The use of radiotracers plays a crucial role in this approach, enabling imaging and therapeutic interventions. Recent advancements in nanobiotechnology have led to the development of nanoparticles functionalized with bisphosphonates, improving the delivery of therapeutic agents directly to osteolytic lesions while facilitating real-time imaging of tumor responses. Additionally, combining PET and magnetic resonance imaging, multimodal imaging techniques provide a comprehensive view of tumor characteristics and treatment efficacy. However, the clinical application of theranostics faces challenges, including regulatory hurdles, high costs, limited availability, and the need for extensive research to demonstrate safety and efficacy. This book chapter explores the innovative approach of theranostics, which integrates diagnostic imaging and targeted therapy to enhance patient outcomes in bone cancer management. This chapter also discusses emerging trends in theranostic technologies, such as investigational radiopharmaceuticals and combination therapies, emphasizing the potential for personalized medicine in bone cancer treatment. Overall, while theranostics holds great promise for improving outcomes in bone cancer, ongoing research is essential to address existing limitations and expand its applications in clinical practice.

Link: https://doi.org/10.1016/B978-0-443-34199-1.00013-2

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Data-driven education: Leveraging big data, AI, and machine learning for smarter learning environments

AUTHOR :

Jain, R.; Upreti, K.; Akilbasha, U.; Radhakrishnan, G.; Sharma, A.; Malik, K.

JOURNAL NAME:

In book: Intelligent Systems for Neurocognition and Human-Robot-Computer Interaction

DETAILS:

Published on November 2025

ABSTRACT:

Big Data, artificial intelligence (AI), and machine learning are transforming education with self-paced learning, precise insights, and automated decisions. The effects these technologies have on education are transformative, especially when it comes to improving student achievement, advancing administrative operations, and personalizing the learning experience, as stated in this chapter. Big Data collects and analyzes immense volumes of data, while AI powers automation, makes predictions, automates evaluations, and enables the possibility of adaptive learning. Chatbots, recommendation engines, and tutoring systems enhance student-focused digital education. Still, integrating these technologies comes with challenges such as privacy and ethical issues, algorithm discrimination, and data security concerns. Some of the new trends are explainable AI (XAI) for ethical decision-making, blockchain and federated learning for privacy-preserving analytic systems, and verifiable credentials. In addition, XR, AI-based virtual laboratories, and neurosymbolic AI will have great consequences on the future learning environment. AI in education offers scalability and inclusivity but demands ethical regulation, governance, and resource management. This chapter recommends sustained effort in research, policy changes, and ethical integration of AI for the best possible use thereof. The education sector, by incorporating human-centered AI approaches, can create a just, sustainable future of learning that is accessible to all students around the world.

Link: https://doi.org/10.1016/B978-0-443-41660-6.00003-X

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Chatbots in health care: AI-based personalization and EHR integration in patient–doctor communication

AUTHOR :

Kuwar, V.; Kumari, P.; Upreti, K.; Gupta, K.; Akilbasha, U.; Bhide, N.

JOURNAL NAME:

In book: Intelligent Systems for Neurocognition and Human-Robot-Computer Interaction

DETAILS:

Published on November 2025

ABSTRACT:

The artificial intelligence (AI)-driven chatbots in healthcare integration revolutionizes patient–provider interactions for real-time support, communication streamline, and patient engagement. These chatbots connected to natural language processing (NLP) and machine learning provide medical queries resolution, chronic condition management, and scheduling appointments. Despite the advancements, there are gaps remain in the chatbot personalization interactions and Electronic Health Records (EHR) seamless integration. Personalization is crucial for satisfied patient and medical advice. EHR integration enables context-aware responses, error reduction, and better healthcare outcomes. This study effectiveness fosters the evaluation of AI-driven chatbots in healthcare communication personalization and potential benefits examination and EHR integration challenges. Using a mixed-methods approach includes sentiment analysis for patient satisfaction sentiments understanding, thematic analysis for key themes and findings from Patient Message, regression analysis for personalization, EHR integration, and patient outcomes understanding, and structural equation modeling (SEM) highlights the personalization and EHR integration impact on patient satisfaction, engagement, and trust in chatbot technology. The findings reinforcing the healthcare providers need to adopt AI-driven solutions and personalized communication priorities and seamless data integration for patient experience improvement and overall healthcare efficiency.

Link: https://doi.org/10.1016/B978-0-443-41660-6.00018-1

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Recent evidence of theranostics and its perspectives on clinical advances and emerging approaches

AUTHOR :

Rawal, K.B.; Kolte, K.; Chand, S.; Sutar, A.; Singh, R.

JOURNAL NAME:

In book: Theranostics in Cancer Management

DETAILS:

Published on October 2025

ABSTRACT:

Theranostics, an emerging field in oncology, integrates diagnostic and therapeutic modalities to enhance precision medicine. This approach utilizes radiopharmaceuticals, nanotechnology, molecular imaging, and biomarkers to diagnose and treat various cancers, including thyroid, neuroendocrine, breast, and prostate cancer. Advanced techniques such as machine learning, artificial intelligence (AI), CRISPR-Cas9, and immunotheranostics are revolutionizing targeted treatment. Despite significant progress, challenges such as regulatory hurdles, complex manufacturing, and high costs limit widespread clinical adoption. Ongoing clinical trials, including those on prostate-specific membrane antigen-targeted therapy and radiopharmaceuticals, demonstrate promising outcomes. Future advancements, particularly in AI-driven diagnostics and personalized treatment strategies, are expected to enhance patient outcomes and healthcare efficiency. This chapter will provide different evidences which theranostics holds to transform cancer care by integrating cutting-edge technology with personalized treatment.

Link: https://doi.org/10.1016/B978-0-443-34199-1.00004-1

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Recent technologies for TNBC management

AUTHOR :

Inamdar, P.R.; Gawade, A.R.; Bhandurge, P.J.; Adsule, P.V.; Kore, P.S.; Kothawade, S.

JOURNAL NAME:

In book: Artificial Intelligence-Driven Precision Medicine for Triple Negative Breast Cancer

DETAILS:

Published on October 2025

ABSTRACT:

Triple-negative breast cancer (TNBC) is the most active area of research in oncology. The name of this cancer was coined on the basis of the absence of expression of three major receptors such as estrogen receptor, progesterone receptor, and human epidermal growth factor receptor 2. This leads to increase in the metastasis and thus creates resistance to chemotherapy. According to recent studies, nearly 20% of the patients with TNBC develop a mutation as breast cancer susceptibility gene (BRCA), causing frequent recurrence with delayed prognosis and treatment. This scenario has restricted the treatment of TNBC to traditional chemotherapy and radiotherapy, and thus, the limited source of treatments and recurrent resistance of TNBC emerged in the development of alternative techniques to alleviate its symptoms and prevalence. This chapter aims to focus on such alternative, dynamic and emerging technologies in the management of TNBC.

Link: https://doi.org/10.1016/B978-0-443-33565-5.00011-6

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Parametric optimization of turning process using PVD coated tools: ant lion algorithm approach

AUTHOR :

Kulkarni, A.; Anerao, P.; Schiffers, C.; Adwani, M.; Kulkarni, O.; Kakandikar, G.

JOURNAL NAME:

Powder Metallurgy: Characterization, Processing, and Optimization Techniques (Book chapter)

DETAILS:

Published on 3 September 2025

ABSTRACT:

Austenitic treated steel has amazing properties like high erosion opposition, high temperature obstruction, and high sturdiness, making it an alluring decision for a wide scope of utilizations. However, it is amongst the “difficult-to-cut” material due to work hardening, high strength, and low thermal conductivity. This paper focuses on experiments that were carried out using multilayer TiAlN/TiSiN-coated tools. The coatings were stored utilizing DC magnetron sputtering (DCMS) PVD procedures on (K-grade) uncoated established carbide embeds. The influence of coating architecture, coating properties, and cutting parameters on the machinability of SS 304 was investigated during dry turning. Feed, speed, and depth of cut are selected as cutting parameters for the investigation and its impact on cutting force. The regression model was developed to predict the values of different response variables and also validated experimentally. Bio-inspired and nature-inspired algorithms play a very crucial role in optimizing the realistic problems. In this research work, ant lion optimizer was used, which predicted a much better result as compared to the experimental output. It was observed that cutting force was affected significantly by feed. However, cutting speed and depth of cut have minimal impact on the cutting force.

Link: https://doi.org/10.1016/B978-0-443-22025-8.00021-8

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

AI-driven biomarker discovery and validation

AUTHOR :

Raut, K.; Chabukswar, A.R.; Jagdale, S.C.; Giri, P.T.; Patil, Y.

JOURNAL NAME:

In book: Artificial Intelligence-Driven Precision Medicine for Triple Negative Breast Cancer

DETAILS:

Published on October 2025

ABSTRACT:

This chapter explores the transformative potential of artificial intelligence in the discovery and validation of biomarkers for triple-negative breast cancer, a highly aggressive and heterogeneous cancer subtype. Advanced AI technologies, combined with multiomics data, can reveal complex biological patterns that traditional methods often miss. The chapter discusses techniques of machine learning, deep learning, and data mining, stressing their application in identifying, validating, and integrating genomic, proteomic, transcriptomic, and metabolomics data to drive precision medicine. Innovations, such as AI-driven liquid biopsy, predictive modeling, and application of cutting-edge technologies such as CRISPR and single-cell analysis, are described for their applications in improving the accuracy of diagnostics, enabling personalization of treatment strategies, and ultimately, improving outcome for patients. Practical solutions to issues of data quality, algorithm bias, and regulatory hurdles are suggested to underscore the need for standardization and collaboration. The impact of AI-driven precision medicine on improving TNBC outcomes cannot be overstated. As AI technologies continue to advance and integrate with emerging scientific methodologies, they will empower clinicians to make more informed decisions tailored to individual patient profiles. This shift toward personalized treatment strategies holds great promise for enhancing survival rates and quality of life for patients with TNBC, marking a significant step forward in the fight against this aggressive cancer subtype.

Link: https://doi.org/10.1016/B978-0-443-33565-5.00001-3

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Chalcones and their derivatives as anticancer agents: Mechanisms and applications

AUTHOR :

Bhandari, S.V.; Pathak, V.; Shitole, A.; Inamdar, P.R.; Dighe, R.; Saundarkar, A.; Shimpi, Y.; Fattepure, K.B.; Lohakare, P.

JOURNAL NAME:

Chalcones and their Derivatives: From Synthesis to Applications (Book Chapter)

DETAILS:

Published on October 2025

ABSTRACT:

Chalcones, a class of natural and synthetic compounds with a common structural motif, have emerged as promising anticancer agents due to their diverse pharmacological properties. This chapter gives an in-depth overview of the mechanisms of action and applications of chalcones and their derivatives in cancer therapy. The first section of the chapter covers the molecular targets of chalcones, including apoptotic induction, cell cycle arrest, and prevention of angiogenesis and metastasis. It then explores the role of chalcones in modulating various signaling pathways involved in the development of cancer, like Nuclear Factor-kappa B (NF-kB) and mammalian target of rapamycin, and NF-κB pathways. Furthermore, the chapter highlights the synergistic effects of chalcones with other chemotherapeutic agents and their potential use in overcoming multidrug resistance in cancer cells. Finally, the chapter explains the difficulties of the present and future directions in the development of chalcones as effective anticancer agents, emphasizing the importance of structure-activity relationship studies and preclinical and clinical evaluations. Furthermore, the chapter discusses the potential of chalcones to overcome drug resistance in cancer cells, making them promising candidates for combination therapy. The chapter also provides insights into the pharmacokinetic properties of chalcones and their derivatives, as well as their bioavailability and toxicity profiles, which are essential considerations for their clinical development. Overall, this chapter underlines the need for more research to fully utilize the promise of chalcones and their derivatives as novel anticancer agents, therapeutic advantages in the management of cancer.

Link: https://doi.org/10.1016/B978-0-443-30122-3.00006-6

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Theranostics in Cancer Management

AUTHOR :

Anitha, K.; Bhatt, S.; Chenchula, S.

JOURNAL NAME:

Book: Theranostics in Cancer Management: Integrating Diagnostics and Therapeutic Strategies

DETAILS:

Published on September 2025

ABSTRACT:

Theranostics in Cancer Management: Integrating Diagnostics and Therapeutic Strategies introduces a transformative paradigm in clinical oncology, heralding a new era of precision and personalized cancer care. This comprehensive guide empowers clinicians by seamlessly integrating advanced diagnostics with targeted therapeutic approaches. The book provides an extensive exploration of theranostics across diverse clinical scenarios, ensuring that healthcare professionals are equipped with the knowledge to navigate the complexities of cancer management effectively. Beyond the introductory insights, the book delves into several types of cancer and the utilization of Artificial Intelligence in cancer detection tools.

Comprising 15 chapters, it offers an invaluable resource for health professionals, scientists, researchers, practitioners, and students who seek to expand their understanding in this evolving field. The emphasis on practical applications and emerging technologies makes it a timely addition to the field of oncology.

Link: https://www.sciencedirect.com/book/edited-volume/9780443341991/theranostics-in-cancer-management

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Introduction to theranostics in oncology

AUTHOR :

Baidya, M.; Keswani, R.; Bhatt, S.

JOURNAL NAME:

In book: Theranostics in Cancer Management

DETAILS:

Published on September 2025

ABSTRACT:

Theranostics is a novel approach that combines therapy and diagnostics. It can be used as personalized approach for the cancer treatment. This integrated approach is used to maximize therapeutic efficacy while reducing their side effects by using treatment to identify and treat various cancer types. Molecular imaging and radionuclide therapy are important for theranostics, as they enable accurate view of tumor as well as delivery of potential chemicals to tumor cells. Most important theranostics are radiolabeled peptides, antibodies, and small molecules, which can be used in conditions like prostate cancer and neuroendocrine tumors. Theranostics enables real-time monitoring of therapy response and adaptive treatment planning by customizing treatment according to the unique molecular features of a tumor. With a focus on theranostics' transformational potential in oncology to enhance outcomes and quality of life for cancer patients, this introduction examines the concepts, applications, challenges, and future directions implications of the field.

Link: https://doi.org/10.1016/B978-0-443-34199-1.00010-7

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Factors impacting stress in financial investment due to the use of artificial intelligence: a social networking analysis approach

AUTHOR :

Bhattacharjee, J.; Singh, R.; Pandey, S.; Sharma, R.

JOURNAL NAME:

AI and Society

DETAILS:

Published on December 2025

ABSTRACT:

The application of artificial intelligence (AI) in financial investments has transformed investment decision-making by providing customized investment solutions and advanced predictive analytics. However, using AI in financial investment is accompanied by critical concerns surrounding data bias, market volatility, lack of transparency, and ethical concerns, resulting in investment-related stress. The present study identifies and examines the important factors impacting stress in financial investment due to the application of artificial intelligence. Using a Delphi technique and Social Network Analysis utilizing UCINET software, nine factors were determined, including perceived loss of control, reduced human interaction, algorithm bias, and over-reliance on AI. This research explored the interrelationships and cascading effect of these identified factors on stress, and key stress-inducing factors like transparency issues, decision fatigue, perceived loss of control, and over-reliance on AI are determined as critical factors affecting investor behavior. The findings highlight the significance of addressing psychological and technological challenges through user education, enhanced transparency, and ethical AI practices to maximize investor trust and confidence. This study offers important implications for policymakers, financial institutions, and academics. By addressing the underlying stressors, stakeholders can design equitable, efficient, and user-friendly AI systems that stimulate a more positive and inclusive investment environment.

Link: https://doi.org/10.1007/s00146-025-02754-4

TITLE:

Enhancing biogas production from lignocellulosic biomass: Challenges, innovations, and sustainability pathways

AUTHOR :

Sharma, S.; Meena, P.K.; Nayak, C.; Burande, C.G.; Shrivastava, A.; Chaturvedi, R.

JOURNAL NAME:

Environmental Progress and Sustainable Energy

DETAILS:

Published on December 2025

ABSTRACT:

Anaerobic digestion (AD) is vital in achieving sustainable waste management and renewable energy goals. This review critically evaluates the recent advancements and bottlenecks in biogas technology, focusing on improving process efficiency, environmental performance, and integration within circular economy frameworks. The analysis reveals that synergistic pretreatment. Approaches—including combined thermal–alkaline and biological techniques—can enhance methane yield by 30%–60% for lignocellulosic and food waste substrates. The co-digestion of organic wastes with microalgae improves process stability, nutrient recycling, and biogas productivity by up to 40%, while supporting carbon sequestration. Regarding upgrading, nanocellulose- and MOF-based membranes offer higher methane purity (>98%) and reduced energy input compared to conventional systems. Despite these technological advancements, challenges persist due to inconsistent operational parameters, inadequate feedstock management, weak policy implementation, and limited public participation. The review emphasizes that standardized monitoring frameworks, decentralized biogas models, and integration with SDG-driven policies are essential to realizing the full potential of AD systems. Overall, the review concludes that targeted innovation and policy synergy can make biogas technology a key enabler of SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities), and SDG 13 (Climate Action).

Link: https://doi.org/10.1002/ep.70251

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Exploring and Evaluating Deep Learning Techniques for Traffic Prediction in Urban Environments

AUTHOR :

Buchade, S.; Sakarkar, G.

JOURNAL NAME:

Transportation Research Record

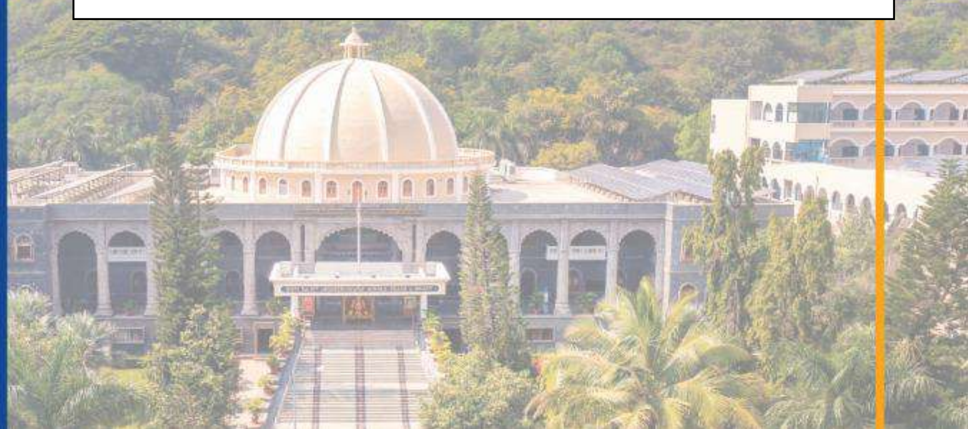
DETAILS:

Published on December 2025

ABSTRACT:

Traffic congestion has been getting worse as a result of the growing population in urban areas that rely on various forms of transportation. However, transportation infrastructure has made significant strides over the last several decades. Traffic prediction plays an important part of intelligent transportation systems in smart cities, promoting relief on traffic congestion. The purpose of this paper is to survey and evaluate deep learning-based traffic forecasting techniques in urban areas. It aims to give a wholesome understanding of how these methods can be used in traffic management and control. The review touches on different methods and mechanisms, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), transformer attention model, graph neural networks or hybrid models such diffusion convolutional recurrent neural networks, spatial-temporal graph neural networks, spatiotemporal graph convolutional networks, and attention-based spatiotemporal graph convolutional networks. This paper also discusses potential real-time applications in traffic prediction, congestion management, and road safety. This is also accompanied with insights on bottlenecks such as data quality, computational constraints, and the need for real-time processing. There are two most effective networks that can handle spatial information—time-series traffic features and complex operations—which are CNNs for short-term prediction and RNNs for long term in the view of deep learning. Their use is particularly evident in smart city initiatives and intelligent transportation systems. Finally, deep learning holds significant promise in this field; however, it needs to overcome these challenges for successful traffic prediction.

Link: https://doi.org/10.1177/03611981251394676



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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

The Social Currency: Unmasking the Amplification of Biases in the Financial World Through Social Media: A Conceptual Review

AUTHOR :

Patil, A.A.; Sengupta, R.; Saha, S.

JOURNAL NAME:

In book: Psychology and Intricacies in Social Media Interactions

DETAILS:

Published on July 2025

ABSTRACT:

Social media platforms are widely used today as a source of information and, consequently, they influence decision-making across nearly every sphere of life, including investment decisions. Behavioral finance is a unique field that combines finance, sociology, and psychology to understand the functioning of financial markets. This article aims to explain how various biases influence investment decisions and how these biases and heuristics are magnified by social media. As a review article, a total of 150 articles were initially retrieved from different databases on the chapter's theme. After scrutiny for relevance, 84 research articles, including newspaper articles, were selected for further review. The review's findings highlight the influence of herding and availability bias, with special attention given to investor attention and social trading platforms. This study finds that social media impacts investment decisions, stock returns, and volatility. Previous research focusing on different artificial intelligence (AI) and machine learning tools applied to social media data and sentiments was also explored. Notably, social media data plays a 208role in predicting and smoothing returns, preventing sudden crashes through the hoarding aversion effect, and influencing investor attention. Given the substantial impact of social media on financial and investment decisions, the authors suggest the need for effective regulation of social media information.

Link: https://doi.org/10.1201/9781003614326-12?urlappend=%3Futm_source%3Dresearchgate.net%26utm_medium%3Darticle

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Molecular glues evolve from serendipity to rational design

AUTHOR :

Shrivastav, P.; Singh, R.; Wiemer, A.J.

JOURNAL NAME:

Trends in Pharmacological Sciences

DETAILS:

Published on December 2025

ABSTRACT:

Molecular glues (MGs) modulate protein interactions to inhibit, activate, or degrade targets. Classical MGs like thalidomide, cyclosporin, and fusicoccin were discovered serendipitously; a key challenge is to develop rational design approaches for MGs. Recent advances in knowledge of the structure-activity relationships (SAR) of MGs have resulted in rational design of new MGs. Moreover, newer structure-based design technologies have increased the target diversity of preclinical MGs and multiple candidates are in clinical trials. Here, we highlight the evolution of MGs from natural products to synthetic compounds and discuss integration of emerging technologies to inform their rational design into new therapeutic agents.

Link: https://doi.org/10.1016/j.tips.2025.10.013

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Water, Weeds, and Warnings: AI at the Frontlines of Modern Farming

AUTHOR :

Pillai, M.R.; Gunjal, A.

JOURNAL NAME:

Conference: 2025 2nd International Conference on Intelligent Algorithms for Computational Intelligence Systems (IACIS)

DETAILS:

Published on November 2025

ABSTRACT:

Agriculture has long struggled to maximize irrigation efficiency and contain crop diseases that damage productivity and sustainability. Conventional irrigation practices generally waste water, and late detection of diseases causes massive losses in crops. Due to rapid developments in artificial intelligence (AI), data-driven responses improved precision agriculture by coupling real-time environmental data, remote sensing technologies, and deep learning models. Using meteorological data and soil moisture content, AI-powered irrigation recommendation systems help distribute water more efficiently, maintaining crop health while reducing excess water use. The Convolutional Neural Network (CNN) makes it possible to detect diseases early by analyzing visual data, providing a chance for earlier treatment. The study assesses the potential of existing AI methods, specifically the use of VGG16, EfficientNet-B0 and MobileNetV2 for supporting agriculture challenges in intelligent automation. MobileNetV2 provided the best overall accuracy of 97% and shows great potential to support agricultural resilience through a lightweight, efficient deep learning solution.

Link: https://doi.org/10.1109/IACIS65746.2025.11211352

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Offboard Control of Autonomous Drones in Ros2-Px4 Simulated Environments

AUTHOR :

Bopalkar, A.; Patil, D.; Somvanshi, H.

JOURNAL NAME:

Conference: 2025 IEEE 4th World Conference on Applied Intelligence and Computing (AIC)

DETAILS:

Published on November 2025

ABSTRACT:

This paper formulates a simulation-driven autonomous drone control framework which utilizes offboard computing for enhanced scalability, adaptability, and efficiency of drone tasks. A crucial advancement within this framework is the separation of control logic and the drone's physical bounds. This makes possible real-time execution of sophisticated multi-layered decisions and detailed control strategies on external systems. The integration of ROS2 middleware with PX4 autopilot, MAVROS bridge, and Gazebo simulator forms a drone control system with modular and extendable architecture. This approach poses higher adaptability and scalability for deployment scenarios compared to other methods especially in computation intensive tasks or large fleet missions. We describe the system architecture and its implementation, and assess system performance using a suite of dynamic simulations that showcase stable, precise flight control. Applications include autonomous drone surveillance, environmental monitoring, and search and rescue missions. This work also aims toward enabling swarm functionalities for autonomous collaboration among multiple drones. This work demonstrated the significant benefits of offboard computing in providing scalable, resilient, and flexible mission adaptable drone systems.

Link: https://doi.org/10.1109/AIC66080.2025.11211887

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॥ विश्वशान्तिर्धुवं ध्रुवा ॥

TITLE:

Sustainable Synergy: Integrating Environmental Economics and Circular Economy

AUTHOR :

Patil, A.A.; Sengupta, R.; Gaurav, K.; Saha, S.; Yadav, R.

JOURNAL NAME:

In book: Circular Economy and Environmental Resilience

DETAILS:

Published on 17 June 2025

ABSTRACT:

The environment has been deteriorated at a rapid pace during the twentieth and twenty-first centuries. Circular economy and sustainability have become popular as possible answers to challenges of the environment and society. The article is based on the relationship between environmental economics and circular economy. Further, the fundamental principles that have to be understood for sustainable development have been explored. According to Environmental Economics, it is crucial to put prices for resources and, also, to include the costs that are given to the environment. Conversely, circular economy signifies a sustainable economic model. It seeks to minimize waste and maximize resource utilization. In this work, authors seek to examine whether it is possible to bridge these two fields and whether the principles of the two fields combined can be a basis for new approaches to tackle the global environmental problems, increase the economic stability and promote sustainable development. Case studies and review of literature have been used to demonstrate how this approach can be the first step to developing sustainable, circular and fair economies and hence assist in the creation of a sustainable and progressive thinking world economy. The findings of the current literature are utilized to develop a broad understanding of the association between environmental economics and circular economy. This has been demonstrated through studying the industrial symbiosis model in Denmark, the energy programme of Sweden, the circular economy model of Steelcase and electric mobility in China. Emphasis has been made on how these two concepts can be integrated so as to achieve sustainable development. Insights of this work will be beneficial to policy makers, organizations and practitioners who are seeking for a sustainable integration of environmental economics and circular economy.

Link: https://doi.org/10.1007/978-3-031-93091-1_10

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Revolutionizing Sustainable Construction with Circuit Board-Derived Concrete: A Comprehensive Review

AUTHOR :

Dudhal, A.; Patil, J.; Minde, P.R.

JOURNAL NAME:

Communications on Applied Nonlinear Analysis (Vol.-32, Issue-1S, 2025)

DETAILS:

Published on September 2025

ABSTRACT:

The increasing amount of waste electronic circuit boards i.e. Printed Circuit Board (PCB) is a serious environmental issue. This chapter explores the ability to use PCB waste as a sustainable addition to concrete, transforming e-waste into a valuable building material. Based on experimental data, the research highlights how adding PCB waste can improve concrete properties, like flexural and compressive strength, while also offering environmental benefits like reduction in usage of natural resources, reduction in landfill waste, and minimizing the carbon footprint of traditional concrete production. The study examines how finely mixed PCB components enhance concrete's strength, durability, and environmental efficiency, showing its suitability for modern infrastructure. It also discusses the role of this innovative material in smart infrastructure, emphasizing its contribution to sustainable development and the promotion of a circular economy. Linking e-waste management with sustainable construction provides a practical solution aligned with global sustainability goals.

Link: https://doi.org/10.1007/978-3-031-92801-7_14

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॥ विश्वशान्तिर्ध्रुवं ध्रुवा ॥

TITLE:

Analytical study of an irregular piezoelectric-orthotropic material substrate subjected to a moving line load

AUTHOR :

Pawar, B.S.; Pal, M.K.; Singh, A.K.

JOURNAL NAME:

JVC/Journal of Vibration and Control

DETAILS:

Published on December 2025

ABSTRACT:

This study analytically investigates the dynamic responses of an Irregular Piezoelectric-Orthotropic Material Substrate (IPOMS) due to a moving line load. The closed-form mathematical expressions of induced Vertical Normal Compressive Stress (VNCS), Transverse Shear Stress (TSS), Horizontal Normal Compressive Stress (HNCS), Vertical Electrical Displacement (VED), and Horizontal Electrical Displacement (HED) has been deduced. Moreover, the influence of various physical parameters, viz., maximum depth of irregularity (ID), irregularity factor (IF), frictional coefficient (FC), and the velocity of a moving line load (VMLL) on said induced electro-mechanical stresses (VNCS, TSS, and HNCS) and dielectric displacements (VED and HED) has been extensively studied. Additionally, a comparative study of IPOMS structure with Irregular Piezoelectric Transversely Isotropic Material Substrate (IPTIMS), Irregular Isotropic Material Substrate (IIMS), and Isotropic Material Substrate (IMS) is conducted and analyzed. Numerical computation and pictorial representation are interpreted with Polyvinylidene difluoride (PVDF), Aluminum nitride (AlN) and isotropic material using MATLAB software.

Link: https://doi.org/10.1177/10775463251399801

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Techno-nationalism and Techno-globalism in Asia: Geopolitical Dynamics and Intellectual Property Implications of Semiconductor Innovations

AUTHOR :

Ghose, A.; Das, M.

JOURNAL NAME:

In book: Decoding the Chessboard of Asian Geopolitics

DETAILS:

Published on 19 April 2025

ABSTRACT:

The semiconductor industry is vital to national security and economic success. A hasty effort to tighten regulations on knowledge transfer and bolster government spending in the semiconductor industry has ensued as a result. In the semiconductor sector, the quest of competitiveness and self-sufficiency has started and will continue to grow. Asia, especially China, has emerged as a prominent center for research and manufacturing in the semiconductor industry, posing a challenge to the traditional supremacy of the United States. This transition has had a profound effect on the geopolitical landscape, which is the primary topic of this study. This study provides a more in-depth examination of the technology gap, specifically concentrating on the progress of Huawei's semiconductor industry. The chapter discusses the notion of techno-nationalism, which refers to a government's efforts to gain control over technology, particularly in relation to national security and economic well-being. This stands in contrast to techno-globalism, which aims to promote the unrestricted movement of ideas and technology across borders. This study employs a doctrinal method to evaluate the aforementioned trends by closely examining intellectual property laws, policies, and ideas in relation to current legislative frameworks and judicial declarations. This study aims to add to the ongoing conversation on strategies that countries might use to both innovate and protect their intellectual property rights in Asia, within the context of a divided global environment.

Link: https://doi.org/10.1007/978-981-96-3073-8_16

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।। विश्वशान्तिर्ध्रुवं ध्रुवा ।।

TITLE:

Three material photonic quasicrystals using extended rauzy fractals

AUTHOR :

Patil, G.U.; Vishwakarma, A.D.; Subramanium, P.; Jaware, T.H.; Barhate, A.A.; Chaudhari, K.J.

JOURNAL NAME:

Journal of Optics (India)

DETAILS:

Published on November 2025

ABSTRACT:

Photonic crystals (PCs) are materials with periodic dielectric structures that create photonic band gaps (PBGs), enabling advanced manipulation of light. Traditional PCs are well established, yet their performance can be sensitive to structural imperfections, motivating the exploration of alternative architectures such as photonic quasicrystals (PQCs). This study explores PQCs that exhibit aperiodic order and unique optical properties, thereby overcoming some of the limitations associated with periodic structures. Utilizing Rauzy fractals, we design three-material photonic quasicrystals (TMPQCs) that leverage their mathematical properties to achieve enhanced optical effects. By employing advanced substitution rules, we generate diverse optical responses and systematically analyze the transmission spectra of TMPQCs across various configurations and generations. Our findings reveal the impact of material arrangement and refractive index contrasts on PBG formation, demonstrating the potential of TMPQCs for innovative photonic applications. This research contributes to a deeper understanding of quasicrystalline materials and paves the way for the design of advanced photonic devices with tailored optical properties.

Link: https://doi.org/10.1007/s12596-025-02982-3

Published on December 2025

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FACULTY PUBLICATION

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